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矢量分析补充

矢量分析是处理场论的最基本的数学工具，对电磁理论的学习极为重要。它本身并不难，只是数学分析课中Green公式、Gauss公式和Stokes公式的延伸。主要就是梯度、散度和旋度的计算，以及几个简单的积分变换公式。在基础物理这个阶段，我们只须学懂这些公式，能看懂书，能理解别人的推理就可以了，尚不需要能够耍弄娴熟。使用的技巧并不多，但也有一些，随着学习的深入，慢慢就会了，不必太着急。下面，我们先复习一下数学课中所讲的梯度、散度、旋度以及Gauss公式和Stokes公式，然后再讲它们的应用与推广，所得的几个公式在物理上都是很有用的。

§0.1 梯度、散度和旋度

设 ψ 为标量场， $\mathbf{f}$ 为矢量场，即

$$\psi = \psi(x, y, z), \quad (0.1.1)$$

$$\mathbf{f} = \mathbf{f}(x, y, z) = \mathbf{i} f_1(x, y, z) + \mathbf{j} f_2(x, y, z) + \mathbf{k} f_3(x, y, z). \quad (0.1.2)$$

在直角坐标下，梯度、散度和旋度的表达式如下：

$$\begin{aligned} \text{grad} \psi &= \nabla \psi \\ &= \mathbf{i} \frac{\partial \psi}{\partial x} + \mathbf{j} \frac{\partial \psi}{\partial y} + \mathbf{k} \frac{\partial \psi}{\partial z}, \end{aligned} \quad (0.1.3)$$

$$\begin{aligned} \text{div} \mathbf{f} &= \nabla \cdot \mathbf{f} \\ &= \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \end{aligned} \quad (0.1.4)$$

$$\begin{aligned} \text{rot} \mathbf{f} &= \nabla \times \mathbf{f}, \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} \\ &= \mathbf{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right). \end{aligned} \quad (0.1.5)$$

它们均是矢量微分算子，

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}, \quad (0.1.6)$$

对场运算的结果。

值得指出的是， ∇ 算子一身二任，既有矢量运算也有偏导运算。不过，在直角坐标下，矢量运算和偏导运算是可以交换的，即

$$\begin{aligned}\nabla &= \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k},\end{aligned}\quad (0.1.7)$$

这是因为三个坐标矢量 $\mathbf{i}, \mathbf{j}, \mathbf{k}$ 均是常矢量。因此，人们可以根据方便先计算矢量运算再计算偏导运算，或先计算偏导运算再计算矢量运算，均可。例如，在对矢量场 $\mathbf{f}$ 取散度和旋度时，方便的办法便是先计算矢量运算——点乘和叉乘，再计算偏导运算。另外，梯度比较简单，只涉及偏导运算。下面的例子则相反，只涉及矢量运算，

$$\mathbf{a} \cdot \nabla = a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}, \quad (0.1.8)$$

其中， $\mathbf{a}$ 是一个常矢量，

$$\mathbf{a} = \mathbf{i} a_1 + \mathbf{j} a_2 + \mathbf{k} a_3. \quad (0.1.9)$$

上述算子 $\mathbf{a} \cdot \nabla$ 也是一个常用的算子，它是一个标量型的微分算子。再如，

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial}{\partial x} \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \frac{\partial}{\partial z} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \quad (0.1.10)$$

有时，也将上式记为

$$\Delta = \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \quad (0.1.11)$$

算子 Δ 称为Laplace算子 (Laplacian)，是Laplace最先引入的，它是一个在数学、物理中常用的算子。本算子也是一个标量算子，不同于上一个标量算子，它含有二阶偏导。这两个标量算子作用在标量场上，结果仍是标量场，作用在矢量场上，结果仍是矢量场：

$$\mathbf{a} \cdot \nabla \psi = a_1 \frac{\partial \psi}{\partial x} + a_2 \frac{\partial \psi}{\partial y} + a_3 \frac{\partial \psi}{\partial z}, \quad (0.1.12)$$

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}, \quad (0.1.13)$$

$$\mathbf{a} \cdot \nabla \mathbf{f} = a_1 \frac{\partial \mathbf{f}}{\partial x} + a_2 \frac{\partial \mathbf{f}}{\partial y} + a_3 \frac{\partial \mathbf{f}}{\partial z}, \quad (0.1.14)$$

$$\nabla^2 \mathbf{f} = \frac{\partial^2 \mathbf{f}}{\partial x^2} + \frac{\partial^2 \mathbf{f}}{\partial y^2} + \frac{\partial^2 \mathbf{f}}{\partial z^2}. \quad (0.1.15)$$

§0.2 高斯公式和斯托克公式

Gauss公式为

$$\oint \mathbf{f} \cdot d\mathbf{s} = \int \nabla \cdot \mathbf{f} dv. \quad (0.2.1)$$

在直角坐标系下,

$$\mathbf{ds} = \mathbf{i} dydz + \mathbf{j} dzdx + \mathbf{k} dxdy, \quad (0.2.2)$$

$$dv = dxdydz, \quad (0.2.3)$$

因此, Gauss公式也可以写为如下形式,

$$\oint f_1 dydz + f_2 dzdx + f_3 dxdy = \int \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) dxdydz. \quad (0.2.4)$$

Stoke公式为

$$\oint \mathbf{f} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{f}) \cdot \mathbf{ds}. \quad (0.2.5)$$

在直角坐标系下,

$$d\mathbf{l} = \mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz, \quad (0.2.6)$$

$$\mathbf{ds} = \mathbf{i} dydz + \mathbf{j} dzdx + \mathbf{k} dxdy, \quad (0.2.7)$$

因此, Stoke公式也可写为如下形式,

$$\begin{aligned} & \oint f_1 dx + f_2 dy + f_3 dz \\ &= \int \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) dydz + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) dzdx + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dxdy. \end{aligned} \quad (0.2.8)$$

标量场的梯度, 矢量场的散度和旋度, 以及Gauss公式和Stokes公式是矢量分析的最基本的公式。下面介绍的内容是它们的应用与推广。

§0.3 乘积场的梯度、散度和旋度

在实际应用中, 经常碰到需要对乘积场取梯度、散度或旋度, 乘积场可以是两个标量场的乘积, 也可是一个标量场与一个矢量场的乘积, 或者两个矢量场的点乘或叉乘。下面的六个公式囊括尽矣, 其中 φ 和 ψ 是标量场, $\mathbf{f}$ 和 $\mathbf{g}$ 是矢量场。

1.

$$\nabla(\varphi\psi) = (\nabla\varphi)\psi + \varphi(\nabla\psi).$$

2.

$$\nabla \cdot (\varphi\mathbf{f}) = (\nabla\varphi) \cdot \mathbf{f} + \varphi(\nabla \cdot \mathbf{f}).$$

3.

$$\nabla \times (\varphi\mathbf{f}) = (\nabla\varphi) \times \mathbf{f} + \varphi(\nabla \times \mathbf{f}).$$

4.

$$\nabla(\mathbf{f} \cdot \mathbf{g}) = \mathbf{f} \times (\nabla \times \mathbf{g}) + (\mathbf{f} \cdot \nabla)\mathbf{g} + \mathbf{g} \times (\nabla \times \mathbf{f}) + (\mathbf{g} \cdot \nabla)\mathbf{f}.$$

5.

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = (\nabla \times \mathbf{f}) \cdot \mathbf{g} - (\nabla \times \mathbf{g}) \cdot \mathbf{f}.$$

6.

$$\nabla \times (\mathbf{f} \times \mathbf{g}) = (\mathbf{g} \cdot \nabla) \mathbf{f} - (\nabla \cdot \mathbf{f}) \mathbf{g} - (\mathbf{f} \cdot \nabla) \mathbf{g} + (\nabla \cdot \mathbf{g}) \mathbf{f}.$$

证明. 我们分别证明如下:

1.

$$\begin{aligned} \nabla(\varphi\psi) &= \mathbf{i} \frac{\partial}{\partial x}(\varphi\psi) + \mathbf{j} \frac{\partial}{\partial y}(\varphi\psi) + \mathbf{k} \frac{\partial}{\partial z}(\varphi\psi) \\ &= \mathbf{i} \left(\frac{\partial \varphi}{\partial x} \psi + \frac{\partial \psi}{\partial x} \varphi \right) + \mathbf{j} \left(\frac{\partial \varphi}{\partial y} \psi + \frac{\partial \psi}{\partial y} \varphi \right) + \mathbf{k} \left(\frac{\partial \varphi}{\partial z} \psi + \frac{\partial \psi}{\partial z} \varphi \right) \\ &= \left(\mathbf{i} \frac{\partial \varphi}{\partial x} \psi + \mathbf{j} \frac{\partial \varphi}{\partial y} \psi + \mathbf{k} \frac{\partial \varphi}{\partial z} \psi \right) + \left(\mathbf{i} \frac{\partial \psi}{\partial x} \varphi + \mathbf{j} \frac{\partial \psi}{\partial y} \varphi + \mathbf{k} \frac{\partial \psi}{\partial z} \varphi \right) \\ &= (\nabla \varphi) \psi + \varphi (\nabla \psi). \end{aligned}$$

2. 令

$$\mathbf{f} = f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k},$$

则

$$\begin{aligned} \nabla \cdot (\varphi \mathbf{f}) &= \frac{\partial}{\partial x}(\varphi f_1) + \frac{\partial}{\partial y}(\varphi f_2) + \frac{\partial}{\partial z}(\varphi f_3) \\ &= \left(\frac{\partial \varphi}{\partial x} f_1 + \frac{\partial f_1}{\partial x} \varphi \right) + \left(\frac{\partial \varphi}{\partial y} f_2 + \frac{\partial f_2}{\partial y} \varphi \right) + \left(\frac{\partial \varphi}{\partial z} f_3 + \frac{\partial f_3}{\partial z} \varphi \right) \\ &= \left(\frac{\partial \varphi}{\partial x} f_1 + \frac{\partial \varphi}{\partial y} f_2 + \frac{\partial \varphi}{\partial z} f_3 \right) + \left(\frac{\partial f_1}{\partial x} \varphi + \frac{\partial f_2}{\partial y} \varphi + \frac{\partial f_3}{\partial z} \varphi \right) \\ &= (\nabla \varphi) \cdot \mathbf{f} + \varphi (\nabla \cdot \mathbf{f}). \end{aligned}$$

3. 同上, 令

$$\mathbf{f} = f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k},$$

我们有

$$\begin{aligned} \nabla \times (\varphi \mathbf{f}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \varphi f_1 & \varphi f_2 & \varphi f_3 \end{vmatrix} \\ &= \mathbf{i} \left[\frac{\partial}{\partial y}(\varphi f_3) - \frac{\partial}{\partial z}(\varphi f_2) \right] + \mathbf{j} \left[\frac{\partial}{\partial z}(\varphi f_1) - \frac{\partial}{\partial x}(\varphi f_3) \right] + \mathbf{k} \left[\frac{\partial}{\partial x}(\varphi f_2) - \frac{\partial}{\partial y}(\varphi f_1) \right] \\ &= \mathbf{i} \left[\frac{\partial \varphi}{\partial y} f_3 + \frac{\partial f_3}{\partial y} \varphi - \frac{\partial \varphi}{\partial z} f_2 - \frac{\partial f_2}{\partial z} \varphi \right] + \mathbf{j} \left[\frac{\partial \varphi}{\partial z} f_1 + \frac{\partial f_1}{\partial z} \varphi - \frac{\partial \varphi}{\partial x} f_3 - \frac{\partial f_3}{\partial x} \varphi \right] \end{aligned}$$

$$\begin{aligned}
& +\mathbf{k} \left[\frac{\partial \varphi}{\partial x} f_2 + \frac{\partial f_2}{\partial x} \varphi - \frac{\partial \varphi}{\partial y} f_1 - \frac{\partial f_1}{\partial y} \varphi \right] \\
= & \left[\mathbf{i} \left(\frac{\partial \varphi}{\partial y} f_3 - \frac{\partial \varphi}{\partial z} f_2 \right) + \mathbf{j} \left(\frac{\partial \varphi}{\partial z} f_1 - \frac{\partial \varphi}{\partial x} f_3 \right) + \mathbf{k} \left(\frac{\partial \varphi}{\partial x} f_2 - \frac{\partial \varphi}{\partial y} f_1 \right) \right] \\
& + \varphi \left[\mathbf{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \right] \\
= & (\nabla \varphi) \times \mathbf{f} + \varphi (\nabla \times \mathbf{f}).
\end{aligned}$$

4. 令

$$\begin{aligned}
\mathbf{f} &= f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}, \\
\mathbf{g} &= g_1 \mathbf{i} + g_2 \mathbf{j} + g_3 \mathbf{k},
\end{aligned}$$

则

$$\begin{aligned}
\nabla(\mathbf{f} \cdot \mathbf{g}) &= \nabla(f_1 g_1 + f_2 g_2 + f_3 g_3) \\
&= \nabla(f_1 g_1) + \nabla(f_2 g_2) + \nabla(f_3 g_3) \\
&= (\nabla f_1) g_1 + (\nabla g_1) f_1 + (\nabla f_2) g_2 + (\nabla g_2) f_2 + (\nabla f_3) g_3 + (\nabla g_3) f_3 \\
&= [f_1 (\nabla g_1) + f_2 (\nabla g_2) + f_3 (\nabla g_3)] + [g_1 (\nabla f_1) + g_2 (\nabla f_2) + g_3 (\nabla f_3)]. \quad (0.3.1)
\end{aligned}$$

又,

$$\begin{aligned}
\mathbf{f} \times (\nabla \times \mathbf{g}) &= \mathbf{f} \times \left[\mathbf{i} \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) \right] \\
&= \mathbf{i} \left[f_2 \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) - f_3 \left(\frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x} \right) \right] \\
&\quad + \mathbf{j} \left[f_3 \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z} \right) - f_1 \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) \right] \\
&\quad + \mathbf{k} \left[f_1 \left(\frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x} \right) - f_2 \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z} \right) \right].
\end{aligned}$$

又,

$$\begin{aligned}
(\mathbf{f} \cdot \nabla) \mathbf{g} &= (\mathbf{f} \cdot \nabla)(\mathbf{i} g_1 + \mathbf{j} g_2 + \mathbf{k} g_3) \\
&= (\mathbf{f} \cdot \nabla)(\mathbf{i} g_1) + (\mathbf{f} \cdot \nabla)(\mathbf{j} g_2) + (\mathbf{f} \cdot \nabla)(\mathbf{k} g_3) \\
&= \mathbf{i} [(\mathbf{f} \cdot \nabla) g_1] + \mathbf{j} [(\mathbf{f} \cdot \nabla) g_2] + \mathbf{k} [(\mathbf{f} \cdot \nabla) g_3].
\end{aligned}$$

注意到

$$(\mathbf{f} \cdot \nabla) g_1 = \left(f_1 \frac{\partial}{\partial x} + f_2 \frac{\partial}{\partial y} + f_3 \frac{\partial}{\partial z} \right) g_1$$

$$\begin{aligned}
&= f_1 \frac{\partial g_1}{\partial x} + f_2 \frac{\partial g_1}{\partial y} + f_3 \frac{\partial g_1}{\partial z}, \\
(\mathbf{f} \cdot \nabla)g_2 &= f_1 \frac{\partial g_2}{\partial x} + f_2 \frac{\partial g_2}{\partial y} + f_3 \frac{\partial g_2}{\partial z}, \\
(\mathbf{f} \cdot \nabla)g_3 &= f_1 \frac{\partial g_3}{\partial x} + f_2 \frac{\partial g_3}{\partial y} + f_3 \frac{\partial g_3}{\partial z},
\end{aligned}$$

我们有

$$\begin{aligned}
(\mathbf{f} \cdot \nabla)\mathbf{g} &= \mathbf{i} \left[f_1 \frac{\partial g_1}{\partial x} + f_2 \frac{\partial g_1}{\partial y} + f_3 \frac{\partial g_1}{\partial z} \right] + \mathbf{j} \left[f_1 \frac{\partial g_2}{\partial x} + f_2 \frac{\partial g_2}{\partial y} + f_3 \frac{\partial g_2}{\partial z} \right] \\
&\quad + \mathbf{k} \left[f_1 \frac{\partial g_3}{\partial x} + f_2 \frac{\partial g_3}{\partial y} + f_3 \frac{\partial g_3}{\partial z} \right].
\end{aligned}$$

由此可知

$$\begin{aligned}
\mathbf{f} \times (\nabla \times \mathbf{g}) + (\mathbf{f} \cdot \nabla)\mathbf{g} &= \mathbf{i} \left[f_1 \frac{\partial g_1}{\partial x} + f_2 \frac{\partial g_2}{\partial x} + f_3 \frac{\partial g_3}{\partial x} \right] \\
&\quad + \mathbf{j} \left[f_1 \frac{\partial g_1}{\partial y} + f_2 \frac{\partial g_2}{\partial y} + f_3 \frac{\partial g_3}{\partial y} \right] \\
&\quad + \mathbf{k} \left[f_1 \frac{\partial g_1}{\partial z} + f_2 \frac{\partial g_2}{\partial z} + f_3 \frac{\partial g_3}{\partial z} \right] \\
&= f_1(\nabla g_1) + f_2(\nabla g_2) + f_3(\nabla g_3).
\end{aligned}$$

同理可得

$$\mathbf{g} \times (\nabla \times \mathbf{f}) + (\mathbf{g} \cdot \nabla)\mathbf{f} = g_1(\nabla f_1) + g_2(\nabla f_2) + g_3(\nabla f_3).$$

将以上两式相加, 并注意到式(0.3.1), 我们得

$$\nabla(\mathbf{f} \cdot \mathbf{g}) = \mathbf{f} \times (\nabla \times \mathbf{g}) + (\mathbf{f} \cdot \nabla)\mathbf{g} + \mathbf{g} \times (\nabla \times \mathbf{f}) + (\mathbf{g} \cdot \nabla)\mathbf{f}.$$

命题成立。

5. 同上, 令

$$\begin{aligned}
\mathbf{f} &= f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}, \\
\mathbf{g} &= g_1 \mathbf{i} + g_2 \mathbf{j} + g_3 \mathbf{k},
\end{aligned}$$

则

$$\mathbf{f} \times \mathbf{g} = \mathbf{i}(f_2 g_3 - f_3 g_2) + \mathbf{j}(f_3 g_1 - f_1 g_3) + \mathbf{k}(f_1 g_2 - f_2 g_1).$$

于是,

$$\begin{aligned}
 \nabla \cdot (\mathbf{f} \times \mathbf{g}) &= \frac{\partial}{\partial x}(f_2g_3 - f_3g_2) + \frac{\partial}{\partial y}(f_3g_1 - f_1g_3) + \frac{\partial}{\partial z}(f_1g_2 - f_2g_1) \\
 &= \left[\frac{\partial f_2}{\partial x}g_3 - \frac{\partial f_3}{\partial x}g_2 + \frac{\partial g_3}{\partial x}f_2 - \frac{\partial g_2}{\partial x}f_3 \right] \\
 &\quad + \left[\frac{\partial f_3}{\partial y}g_1 - \frac{\partial f_1}{\partial y}g_3 + \frac{\partial g_1}{\partial y}f_3 - \frac{\partial g_3}{\partial y}f_1 \right] \\
 &\quad + \left[\frac{\partial f_1}{\partial z}g_2 - \frac{\partial f_2}{\partial z}g_1 + \frac{\partial g_2}{\partial z}f_1 - \frac{\partial g_1}{\partial z}f_2 \right] \\
 &= \left[\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) g_1 + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) g_2 + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) g_3 \right] \\
 &\quad - \left[\left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z} \right) f_1 + \left(\frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x} \right) f_2 + \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) f_3 \right] \\
 &= (\nabla \times \mathbf{f}) \cdot \mathbf{g} - (\nabla \times \mathbf{g}) \cdot \mathbf{f}.
 \end{aligned}$$

6. 同上, 令

$$\begin{aligned}
 \mathbf{f} &= f_1 \mathbf{i} + f_2 \mathbf{j} + f_3 \mathbf{k}, \\
 \mathbf{g} &= g_1 \mathbf{i} + g_2 \mathbf{j} + g_3 \mathbf{k},
 \end{aligned}$$

则有

$$\mathbf{f} \times \mathbf{g} = \mathbf{i}(f_2g_3 - f_3g_2) + \mathbf{j}(f_3g_1 - f_1g_3) + \mathbf{k}(f_1g_2 - f_2g_1).$$

于是,

$$\begin{aligned}
 \nabla \times (\mathbf{f} \times \mathbf{g}) &= \mathbf{i} \left[\frac{\partial}{\partial y}(f_1g_2 - f_2g_1) - \frac{\partial}{\partial z}(f_3g_1 - f_1g_3) \right] \\
 &\quad + \mathbf{j} \left[\frac{\partial}{\partial z}(f_2g_3 - f_3g_2) - \frac{\partial}{\partial x}(f_1g_2 - f_2g_1) \right] \\
 &\quad + \mathbf{k} \left[\frac{\partial}{\partial x}(f_3g_1 - f_1g_3) - \frac{\partial}{\partial y}(f_2g_3 - f_3g_2) \right].
 \end{aligned}$$

下面, 我们来看矢量

$$(\mathbf{g} \cdot \nabla)\mathbf{f} - (\nabla \cdot \mathbf{f})\mathbf{g} - (\mathbf{f} \cdot \nabla)\mathbf{g} + (\nabla \cdot \mathbf{g})\mathbf{f},$$

它的 x 分量等于

$$\begin{aligned}
 &(\mathbf{g} \cdot \nabla)f_1 - (\nabla \cdot \mathbf{f})g_1 - (\mathbf{f} \cdot \nabla)g_1 + (\nabla \cdot \mathbf{g})f_1 \\
 &= \left(g_1 \frac{\partial}{\partial x} + g_2 \frac{\partial}{\partial y} + g_3 \frac{\partial}{\partial z} \right) f_1 - \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) g_1 \\
 &\quad - \left(f_1 \frac{\partial}{\partial x} + f_2 \frac{\partial}{\partial y} + f_3 \frac{\partial}{\partial z} \right) g_1 + \left(\frac{\partial g_1}{\partial x} + \frac{\partial g_2}{\partial y} + \frac{\partial g_3}{\partial z} \right) f_1 \\
 &= \left(g_1 \frac{\partial f_1}{\partial x} + g_2 \frac{\partial f_1}{\partial y} + g_3 \frac{\partial f_1}{\partial z} \right) - \left(g_1 \frac{\partial f_1}{\partial x} + g_1 \frac{\partial f_2}{\partial y} + g_1 \frac{\partial f_3}{\partial z} \right)
 \end{aligned}$$

$$-\left(f_1 \frac{\partial g_1}{\partial x} + f_2 \frac{\partial g_1}{\partial y} + f_3 \frac{\partial g_1}{\partial z}\right) + \left(f_1 \frac{\partial g_1}{\partial x} + f_1 \frac{\partial g_2}{\partial y} + f_1 \frac{\partial g_3}{\partial z}\right).$$

容易看出，上式中关于 x 的偏导项相互抵消了，只剩下关于 y, z 的偏导项。这样，我们有

$$\begin{aligned} & (\mathbf{g} \cdot \nabla) f_1 - (\nabla \cdot \mathbf{f}) g_1 - (\mathbf{f} \cdot \nabla) g_1 + (\nabla \cdot \mathbf{g}) f_1 \\ &= \left[g_2 \frac{\partial f_1}{\partial y} - g_1 \frac{\partial f_2}{\partial y} - f_2 \frac{\partial g_1}{\partial y} + f_1 \frac{\partial g_2}{\partial y} \right] \\ & \quad + \left[g_3 \frac{\partial f_1}{\partial z} - g_1 \frac{\partial f_3}{\partial z} - f_3 \frac{\partial g_1}{\partial z} + f_1 \frac{\partial g_3}{\partial z} \right] \\ &= \frac{\partial}{\partial y} (f_1 g_2 - f_2 g_1) - \frac{\partial}{\partial z} (f_3 g_1 - f_1 g_3). \end{aligned}$$

可见，矢量

$$\nabla \times (\mathbf{f} \times \mathbf{g})$$

与矢量

$$(\mathbf{g} \cdot \nabla) \mathbf{f} - (\nabla \cdot \mathbf{f}) \mathbf{g} - (\mathbf{f} \cdot \nabla) \mathbf{g} + (\nabla \cdot \mathbf{g}) \mathbf{f}$$

的 x 分量相等，同样可证它们的 y, z 分量也分别相等，因此，二者相等，

$$\nabla \times (\mathbf{f} \times \mathbf{g}) = (\mathbf{g} \cdot \nabla) \mathbf{f} - (\nabla \cdot \mathbf{f}) \mathbf{g} - (\mathbf{f} \cdot \nabla) \mathbf{g} + (\nabla \cdot \mathbf{g}) \mathbf{f}.$$

命题成立。 ■

值得指出的是，要快速地导出甚至记住这些公式，只须改变一下对乘积偏导的看法即可，

$$\frac{\partial}{\partial x}(fg) = f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x}.$$

函数乘积 fg 的偏导可以看作：先保持 f 不变，对 g 偏导，再保持 g 不变，对 f 偏导，即

$$\frac{\partial}{\partial x}(fg) = \frac{\partial}{\partial x}(f_c g + f g_c) = \frac{\partial}{\partial x}(f_c g) + \frac{\partial}{\partial x}(f g_c) = f_c \frac{\partial g}{\partial x} + g_c \frac{\partial f}{\partial x}.$$

这里，为了方便，我们特地加了助记符——下标 c ，来提醒在偏导过程中保持不变的函数。最后，去掉助记符，我们就得到熟知的结果，

$$\frac{\partial}{\partial x}(fg) = f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x}.$$

上面那六个矢量分析的公式，都是对乘积的偏导，我们可以按上面的方式处理。例如 $\nabla \cdot (\mathbf{f} \times \mathbf{g})$ ，我们可以先将它写为

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = \nabla \cdot (\mathbf{f}_c \times \mathbf{g} + \mathbf{f} \times \mathbf{g}_c) = \nabla \cdot (\mathbf{f}_c \times \mathbf{g}) + \nabla \cdot (\mathbf{f} \times \mathbf{g}_c).$$

剩下的就是想办法将 $\mathbf{f}_c$ 和 $\mathbf{g}_c$ 搬出偏导运算之外，这就需要使用矢量的代数运算了（记住，此时矢量的分析运算——偏导——已经完成）。利用混合积，

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}),$$

我们有

$$\begin{aligned}\nabla \cdot (\mathbf{f}_c \times \mathbf{g}) &= -\nabla \cdot (\mathbf{g} \times \mathbf{f}_c) = -(\nabla \times \mathbf{g}) \cdot \mathbf{f}_c, \\ \nabla \cdot (\mathbf{f} \times \mathbf{g}_c) &= (\nabla \times \mathbf{f}) \cdot \mathbf{g}_c.\end{aligned}$$

因此,

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = -(\nabla \times \mathbf{g}) \cdot \mathbf{f}_c + (\nabla \times \mathbf{f}) \cdot \mathbf{g}_c.$$

最后, 去掉助记符, 得

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = (\nabla \times \mathbf{f}) \cdot \mathbf{g} - (\nabla \times \mathbf{g}) \cdot \mathbf{f}.$$

它正是我们前面证得的结果。其它的类此。

§0.4 梯度的旋度与旋度的散度

有关梯度、散度和旋度, 有两个及其重要的结果。

1. 梯度的旋度恒为零,

$$\nabla \times (\nabla \psi) = 0. \quad (0.4.1)$$

2. 旋度的散度恒为零,

$$\nabla \cdot (\nabla \times \mathbf{f}) = 0. \quad (0.4.2)$$

证明. 1. 注意到

$$\nabla \psi = \mathbf{i} \frac{\partial \psi}{\partial x} + \mathbf{j} \frac{\partial \psi}{\partial y} + \mathbf{k} \frac{\partial \psi}{\partial z}, \quad (0.4.3)$$

我们有

$$\begin{aligned}\nabla \times (\nabla \psi) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \psi}{\partial x} & \frac{\partial \psi}{\partial y} & \frac{\partial \psi}{\partial z} \end{vmatrix} \\ &= \mathbf{i} \left(\frac{\partial^2 \psi}{\partial y \partial z} - \frac{\partial^2 \psi}{\partial z \partial y} \right) + \mathbf{j} \left(\frac{\partial^2 \psi}{\partial z \partial x} - \frac{\partial^2 \psi}{\partial x \partial z} \right) + \mathbf{k} \left(\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \right) \\ &= 0. \end{aligned} \quad (0.4.4)$$

式(0.4.1)得证。

2. 由于

$$\nabla \times \mathbf{f} = \mathbf{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right), \quad (0.4.5)$$

因此,

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{f}) &= \frac{\partial}{\partial x} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \\ &= \frac{\partial^2 f_3}{\partial x \partial y} - \frac{\partial^2 f_2}{\partial x \partial z} + \frac{\partial^2 f_1}{\partial y \partial z} - \frac{\partial^2 f_3}{\partial x \partial y} + \frac{\partial^2 f_2}{\partial x \partial z} - \frac{\partial^2 f_1}{\partial y \partial z} \\ &= 0. \end{aligned} \quad (0.4.6)$$

式(0.4.2)得证。 ■

§0.5 矢量场的双旋度

因为矢量场的旋度仍然是矢量场，所以还可以再一次取旋度，这里谓之双旋度是也。我们有

$$\nabla \times (\nabla \times \mathbf{f}) = \nabla(\nabla \cdot \mathbf{f}) - \nabla^2 \mathbf{f}. \quad (0.5.1)$$

证明. 首先,

$$\nabla \times \mathbf{f} = \mathbf{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right). \quad (0.5.2)$$

其次,

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{f}) &= \mathbf{i} \left[\frac{\partial}{\partial y} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \right] \\ &\quad + \mathbf{j} \left[\frac{\partial}{\partial z} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) - \frac{\partial}{\partial x} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \right] \\ &\quad + \mathbf{k} \left[\frac{\partial}{\partial x} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \right] \\ &= \mathbf{i} \left[\frac{\partial}{\partial x} \left(\frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \left(\frac{\partial^2 f_1}{\partial y^2} + \frac{\partial^2 f_1}{\partial z^2} \right) \right] \\ &\quad + \mathbf{j} \left[\frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_3}{\partial z} \right) - \left(\frac{\partial^2 f_2}{\partial x^2} + \frac{\partial^2 f_2}{\partial z^2} \right) \right] \\ &\quad + \mathbf{k} \left[\frac{\partial}{\partial z} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} \right) - \left(\frac{\partial^2 f_3}{\partial x^2} + \frac{\partial^2 f_3}{\partial y^2} \right) \right] \\ &= \mathbf{i} \left[\frac{\partial}{\partial x} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \left(\frac{\partial^2 f_1}{\partial x^2} + \frac{\partial^2 f_1}{\partial y^2} + \frac{\partial^2 f_1}{\partial z^2} \right) \right] \\ &\quad + \mathbf{j} \left[\frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \left(\frac{\partial^2 f_2}{\partial x^2} + \frac{\partial^2 f_2}{\partial y^2} + \frac{\partial^2 f_2}{\partial z^2} \right) \right] \\ &\quad + \mathbf{k} \left[\frac{\partial}{\partial z} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \left(\frac{\partial^2 f_3}{\partial x^2} + \frac{\partial^2 f_3}{\partial y^2} + \frac{\partial^2 f_3}{\partial z^2} \right) \right] \\ &= \mathbf{i} \left[\frac{\partial}{\partial x} (\nabla \cdot \mathbf{f}) - \nabla^2 f_1 \right] + \mathbf{j} \left[\frac{\partial}{\partial y} (\nabla \cdot \mathbf{f}) - \nabla^2 f_2 \right] \\ &\quad + \mathbf{k} \left[\frac{\partial}{\partial z} (\nabla \cdot \mathbf{f}) - \nabla^2 f_3 \right] \\ &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (\nabla \cdot \mathbf{f}) - \nabla^2 (\mathbf{i} f_1 + \mathbf{j} f_2 + \mathbf{k} f_3) \\ &= \nabla(\nabla \cdot \mathbf{f}) - \nabla^2 \mathbf{f}. \end{aligned} \quad (0.5.3)$$

在倒数第二个等式中, 我们交换了 *Laplacian* 和三个坐标基矢 $\mathbf{i}, \mathbf{j}, \mathbf{k}$ 的次序, 即

$$\mathbf{i} \nabla^2 f_1 = \nabla^2 (\mathbf{i} f_1), \quad \mathbf{j} \nabla^2 f_2 = \nabla^2 (\mathbf{j} f_2), \quad \mathbf{k} \nabla^2 f_3 = \nabla^2 (\mathbf{k} f_3), \quad (0.5.4)$$

这是因为 $\mathbf{i}, \mathbf{j}, \mathbf{k}$ 均是常矢量。

式(0.5.1)得证。 ■

§0.6 积分变换

在电磁理论中，经常用到下面几个积分变换：

1. $dv \nabla \rightarrow ds$

$$\begin{aligned}\int_V dv \nabla \cdot \mathbf{f} &= \oint_S ds \cdot \mathbf{f}, \\ \int_V dv \nabla \times \mathbf{f} &= \oint_S ds \times \mathbf{f}, \\ \int_V dv \nabla \varphi &= \oint_S ds \varphi.\end{aligned}$$

2. $ds \times \nabla \rightarrow dl$

$$\begin{aligned}\int_S (ds \times \nabla) \cdot \mathbf{f} &= \oint_L dl \cdot \mathbf{f}, \\ \int_S (ds \times \nabla) \times \mathbf{f} &= \oint_L dl \times \mathbf{f}, \\ \int_S ds \times \nabla \varphi &= \oint_L dl \varphi.\end{aligned}$$

3. $dl \cdot \nabla \rightarrow d$

$$\begin{aligned}\int_L dl \cdot \nabla \varphi &= \int_L d\varphi = \int_{\mathbf{r}_1}^{\mathbf{r}_2} d\varphi = \varphi(\mathbf{r}_2) - \varphi(\mathbf{r}_1), \\ \int_L (dl \cdot \nabla) \mathbf{f} &= \int_L d\mathbf{f} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} d\mathbf{f} = \mathbf{f}(\mathbf{r}_2) - \mathbf{f}(\mathbf{r}_1),\end{aligned}$$

证明。 我们分别证明如下：

1. 公式

$$\int_V dv \nabla \cdot \mathbf{f} = \oint_S ds \cdot \mathbf{f}$$

其实就是高斯公式，

$$\oint_S \mathbf{f} \cdot ds = \int_V (\nabla \cdot \mathbf{f}) dv.$$

至于公式

$$\int_V dv \nabla \times \mathbf{f} = \oint_S ds \times \mathbf{f},$$

其左边可写为

$$\begin{aligned} & \int_V dv \nabla \times \mathbf{f} \\ &= \int_V dv \left[\mathbf{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \mathbf{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \mathbf{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \right] \\ &= \mathbf{i} \int_V \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) dv + \mathbf{j} \int_V \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) dv + \mathbf{k} \int_V \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dv. \end{aligned}$$

其右边可写为

$$\begin{aligned} \oint_S \mathbf{ds} \times \mathbf{f} &= \oint_S [\mathbf{i} dy dz + \mathbf{j} dz dx + \mathbf{k} dx dy] \times [\mathbf{i} f_1 + \mathbf{j} f_2 + \mathbf{k} f_3] \\ &= \mathbf{i} \oint_S f_3 dz dx - f_2 dx dy + \mathbf{j} \oint_S f_1 dx dy - f_3 dy dz \\ &\quad + \mathbf{k} \oint_S f_2 dy dz - f_1 dz dx. \end{aligned}$$

我们只须证明两边的三个分量分别相等即可, 例如 x 分量,

$$\oint_S f_3 dz dx - f_2 dx dy = \int_V \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) dv.$$

欲证此式, 只须用高斯公式即可, 也就是,

$$\oint_S g_1 dy dz + g_2 dz dx + g_3 dx dy = \int_V \left(\frac{\partial g_1}{\partial x} + \frac{\partial g_2}{\partial y} + \frac{\partial g_3}{\partial z} \right) dv.$$

此处取 $g_1 = 0$, $g_2 = f_3$, $g_3 = -f_2$, 即可。其它的两个分量可以同样处理。可见, 本公式实际上包含了三个高斯公式, 每分量一个。

最后, 公式

$$\int_V dv \nabla \varphi = \oint_S \mathbf{ds} \varphi$$

可以写为

$$\int_V \left(\mathbf{i} \frac{\partial \varphi}{\partial x} + \mathbf{j} \frac{\partial \varphi}{\partial y} + \mathbf{k} \frac{\partial \varphi}{\partial z} \right) dv = \oint_S (\mathbf{i} \varphi dy dz + \mathbf{j} \varphi dz dx + \mathbf{k} \varphi dx dy)$$

按三个分量, 分别证明它们相等即可。例如 x 分量,

$$\oint_S \varphi dy dz = \int_V \frac{\partial \varphi}{\partial x} dv.$$

明显地, 此公式正是高斯公式的结果: $g_1 = \varphi$, $g_2 = 0$, $g_3 = 0$ 。同前例, 本公式也包含三个高斯公式, 每分量一个。

2. 注意到

$$(\mathbf{ds} \times \nabla) \cdot \mathbf{f} = \mathbf{ds} \cdot (\nabla \times \mathbf{f}),$$

公式

$$\int_S (\mathbf{ds} \times \nabla) \cdot \mathbf{f} = \oint_L \mathbf{dl} \cdot \mathbf{f}$$

其实就是 *Stokes* 公式,

$$\oint_L \mathbf{f} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{f}) \cdot d\mathbf{s}$$

至于第二个公式,

$$\int_S (d\mathbf{s} \times \nabla) \times \mathbf{f} = \oint_L d\mathbf{l} \times \mathbf{f},$$

同前一样, 我们打算按直角坐标分量证之。首先,

$$\begin{aligned} d\mathbf{s} \times \nabla &= (\mathbf{i} dy dz + \mathbf{j} dz dx + \mathbf{k} dx dy) \times \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \\ &= \mathbf{i} \left(dz dx \frac{\partial}{\partial z} - dx dy \frac{\partial}{\partial y} \right) + \mathbf{j} \left(dx dy \frac{\partial}{\partial x} - dy dz \frac{\partial}{\partial z} \right) \\ &\quad + \mathbf{k} \left(dy dz \frac{\partial}{\partial y} - dz dx \frac{\partial}{\partial x} \right). \end{aligned}$$

其次,

$$\begin{aligned} (d\mathbf{s} \times \nabla) \times \mathbf{f} &= \mathbf{i} \left[\left(dx dy \frac{\partial f_3}{\partial x} - dy dz \frac{\partial f_3}{\partial z} \right) - \left(dy dz \frac{\partial f_2}{\partial y} - dz dx \frac{\partial f_2}{\partial x} \right) \right] \\ &\quad + \mathbf{j} \left[\left(dy dz \frac{\partial f_1}{\partial y} - dz dx \frac{\partial f_1}{\partial x} \right) - \left(dz dx \frac{\partial f_3}{\partial z} - dx dy \frac{\partial f_3}{\partial y} \right) \right] \\ &\quad + \mathbf{k} \left[\left(dz dx \frac{\partial f_2}{\partial z} - dx dy \frac{\partial f_2}{\partial y} \right) - \left(dx dy \frac{\partial f_1}{\partial x} - dy dz \frac{\partial f_1}{\partial z} \right) \right]. \end{aligned}$$

再次,

$$\begin{aligned} d\mathbf{l} \times \mathbf{f} &= (\mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz) \times (\mathbf{i} f_1 + \mathbf{j} f_2 + \mathbf{k} f_3) \\ &= \mathbf{i} (f_3 dy - f_2 dz) + \mathbf{j} (f_1 dz - f_3 dx) + \mathbf{k} (f_2 dx - f_1 dy). \end{aligned}$$

这样, 对于 x 分量, 我们只须证明下式即可,

$$\oint_L f_3 dy - f_2 dz = \int_S \left(dx dy \frac{\partial f_3}{\partial x} - dy dz \frac{\partial f_3}{\partial z} \right) - \left(dy dz \frac{\partial f_2}{\partial y} - dz dx \frac{\partial f_2}{\partial x} \right).$$

欲证此式, 只须引用 *Stokes* 公式,

$$\begin{aligned} &\oint g_1 dx + g_2 dy + g_3 dz \\ &= \int \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z} \right) dy dz + \left(\frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x} \right) dz dx + \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) dx dy, \end{aligned}$$

并令 $g_1 = 0$, $g_2 = f_3$, $g_3 = -f_2$ 即可。至于 y 分量和 z 分量, 可以同样地证明, 因此, 本公式实际上包含了三个 *Stokes* 公式, 每分量一个。

最后, 第三个公式,

$$\int_S d\mathbf{s} \times \nabla \varphi = \oint_L d\mathbf{l} \varphi.$$

在直角坐标下, 我们有

$$\begin{aligned}
 d\mathbf{l} \varphi &= (\mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz) \varphi \\
 &= \mathbf{i} \varphi dx + \mathbf{j} \varphi dy + \mathbf{k} \varphi dz, \\
 d\mathbf{s} \times \nabla \varphi &= (\mathbf{i} dy dz + \mathbf{j} dz dx + \mathbf{k} dx dy) \times \left(\mathbf{i} \frac{\partial \varphi}{\partial x} + \mathbf{j} \frac{\partial \varphi}{\partial y} + \mathbf{k} \frac{\partial \varphi}{\partial z} \right) \\
 &= \mathbf{i} \left(\frac{\partial \varphi}{\partial z} dz dx - \frac{\partial \varphi}{\partial y} dx dy \right) + \mathbf{j} \left(\frac{\partial \varphi}{\partial x} dx dy - \frac{\partial \varphi}{\partial z} dy dz \right) \\
 &\quad + \mathbf{k} \left(\frac{\partial \varphi}{\partial y} dy dz - \frac{\partial \varphi}{\partial x} dz dx \right),
 \end{aligned}$$

对于 x 分量, 只须证明

$$\oint_L \varphi dx = \int_S \frac{\partial \varphi}{\partial z} dz dx - \frac{\partial \varphi}{\partial y} dx dy.$$

它显然就是 *Stokes* 公式的结果: $g_1 = \varphi$, $g_2 = 0$, $g_3 = 0$ 。 y 分量和 z 分量类此。可见, 此公式也包含三个 *Stokes* 公式, 每分量一个。

3. 注意到

$$d\mathbf{l} \cdot \nabla = dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} + dz \frac{\partial}{\partial z},$$

我们有

$$\begin{aligned}
 d\mathbf{l} \cdot \nabla f &= dx \frac{\partial f}{\partial x} + dy \frac{\partial f}{\partial y} + dz \frac{\partial f}{\partial z} \\
 &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\
 &= df, \\
 (d\mathbf{l} \cdot \nabla) \mathbf{f} &= (d\mathbf{l} \cdot \nabla)(\mathbf{i} f_1 + \mathbf{j} f_2 + \mathbf{k} f_3) \\
 &= (d\mathbf{l} \cdot \nabla)(\mathbf{i} f_1) + (d\mathbf{l} \cdot \nabla)(\mathbf{j} f_2) + (d\mathbf{l} \cdot \nabla)(\mathbf{k} f_3) \\
 &= \mathbf{i} (d\mathbf{l} \cdot \nabla f_1) + \mathbf{j} (d\mathbf{l} \cdot \nabla f_2) + \mathbf{k} (d\mathbf{l} \cdot \nabla f_3) \\
 &= \mathbf{i} df_1 + \mathbf{j} df_2 + \mathbf{k} df_3 \\
 &= d\mathbf{f}.
 \end{aligned}$$

由上面这两个结果, 本组的两个积分变换公式就很明显了。

■

§0.7 格林第一定理和格林第二定理

- 格林第一定理 (Green's first law):

$$\oint \varphi \nabla \psi \cdot d\mathbf{s} = \int (\varphi \nabla^2 \psi + \nabla \varphi \cdot \nabla \psi) dv. \quad (0.7.1)$$

- 格林第二定理 (Green's first law):

$$\oint (\varphi \nabla \psi - \psi \nabla \varphi) \cdot d\mathbf{s} = \int (\varphi \nabla^2 \psi - \psi \nabla^2 \varphi) dv. \quad (0.7.2)$$

它们都是Gauss公式的推论, 这是因为

$$\begin{aligned} \nabla \cdot (\varphi \nabla \psi) &= (\nabla \varphi) \cdot \nabla \psi + \varphi \nabla \cdot (\nabla \psi) \\ &= \nabla \varphi \cdot \nabla \psi + \varphi \nabla^2 \psi \\ &= \varphi \nabla^2 \psi + \nabla \varphi \cdot \nabla \psi, \end{aligned}$$

以及

$$\begin{aligned} \nabla \cdot (\varphi \nabla \psi - \psi \nabla \varphi) &= \nabla \cdot (\varphi \nabla \psi) - \nabla \cdot (\psi \nabla \varphi) \\ &= \varphi \nabla^2 \psi + \nabla \varphi \cdot \nabla \psi - (\psi \nabla^2 \varphi + \nabla \psi \cdot \nabla \varphi) \\ &= \varphi \nabla^2 \psi - \psi \nabla^2 \varphi. \end{aligned}$$

有时, 人们也将它们也写为以下形式,

$$\oint \varphi \frac{d\psi}{dn} ds = \int (\varphi \nabla^2 \psi + \nabla \varphi \cdot \nabla \psi) dv, \quad (0.7.3)$$

$$\oint \left(\varphi \frac{d\psi}{dn} - \psi \frac{d\varphi}{dn} \right) ds = \int (\varphi \nabla^2 \psi - \psi \nabla^2 \varphi) dv, \quad (0.7.4)$$

其中,

$$\frac{d\psi}{dn} = \mathbf{n} \cdot \nabla \psi, \quad \frac{d\varphi}{dn} = \mathbf{n} \cdot \nabla \varphi, \quad (0.7.5)$$

是沿曲面法矢方向 $\mathbf{n}$ 的方向导数,

$$\mathbf{n} = \frac{d\mathbf{s}}{ds} = \frac{d\mathbf{s}}{ds}. \quad (0.7.6)$$

§0.8 Taylor公式

按照多元函数的Taylor公式, 对标量场 ψ , 我们有

$$\begin{aligned} \psi(\mathbf{r} + \mathbf{d}) &= \psi(\mathbf{r}) + \left(d_1 \frac{\partial}{\partial x} + d_2 \frac{\partial}{\partial y} + d_3 \frac{\partial}{\partial z} \right) \psi(\mathbf{r}) \\ &\quad + \frac{1}{2} \left(d_1 \frac{\partial}{\partial x} + d_2 \frac{\partial}{\partial y} + d_3 \frac{\partial}{\partial z} \right)^2 \psi(\mathbf{r}) + \cdots \\ &\quad + \frac{1}{n!} \left(d_1 \frac{\partial}{\partial x} + d_2 \frac{\partial}{\partial y} + d_3 \frac{\partial}{\partial z} \right)^n \psi(\mathbf{r}) + o(|\mathbf{d}|^n), \end{aligned} \quad (0.8.1)$$

其中, d_1, d_2, d_3 是矢量 $\mathbf{d}$ 的直角坐标,

$$\mathbf{d} = \mathbf{i} d_1 + \mathbf{j} d_2 + \mathbf{k} d_3. \quad (0.8.2)$$

注意到

$$d_1 \frac{\partial}{\partial x} + d_2 \frac{\partial}{\partial y} + d_3 \frac{\partial}{\partial z} = \mathbf{d} \cdot \nabla, \quad (0.8.3)$$

上述Taylor公式还可写为如下形式,

$$\begin{aligned} \psi(\mathbf{r} + \mathbf{d}) &= \psi(\mathbf{r}) + (\mathbf{d} \cdot \nabla)\psi(\mathbf{r}) + \frac{1}{2}(\mathbf{d} \cdot \nabla)^2\psi(\mathbf{r}) + \cdots \\ &\quad + \frac{1}{n!}(\mathbf{d} \cdot \nabla)^n\psi(\mathbf{r}) + o(|\mathbf{d}|^n), \end{aligned} \quad (0.8.4)$$

或者,

$$\psi(\mathbf{r} + \mathbf{d}) = \sum_{k=0}^n \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k\psi(\mathbf{r}) + o(|\mathbf{d}|^n). \quad (0.8.5)$$

如果函数 ψ 在点 $\mathbf{r}$ 解析, 且 $|\mathbf{d}|$ 小于此点的收敛半径, 则

$$\psi(\mathbf{r} + \mathbf{d}) = \sum_{k=0}^{+\infty} \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k\psi(\mathbf{r}). \quad (0.8.6)$$

也就是说, 此时函数 ψ 可以展为Taylor级数。考虑到

$$e^x = \sum_{k=0}^{+\infty} \frac{1}{k!}x^k, \quad (0.8.7)$$

我们可以定义算子的指数函数,

$$e^{\mathbf{d} \cdot \nabla} := \sum_{k=0}^{+\infty} \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k. \quad (0.8.8)$$

于是, 上述Taylor级数可以写为

$$\psi(\mathbf{r} + \mathbf{d}) = e^{\mathbf{d} \cdot \nabla}\psi(\mathbf{r}). \quad (0.8.9)$$

对于矢量场 $\mathbf{f}$,

$$\mathbf{f} = \mathbf{i} f_1(x, y, z) + \mathbf{j} f_2(x, y, z) + \mathbf{k} f_3(x, y, z). \quad (0.8.10)$$

我们可以分别对多它的三个分量进行Taylor展开,

$$f_i(\mathbf{r} + \mathbf{d}) = \sum_{k=0}^n \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k f_i(\mathbf{r}) + o(|\mathbf{d}|^n), \quad i = 1, 2, 3. \quad (0.8.11)$$

上式也可以写为

$$\mathbf{f}(\mathbf{r} + \mathbf{d}) = \sum_{k=0}^n \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k \mathbf{f}(\mathbf{r}) + o(|\mathbf{d}|^n), \quad i = 1, 2, 3. \quad (0.8.12)$$

在解析点附近, 我们有

$$\mathbf{f}(\mathbf{r} + \mathbf{d}) = \sum_{k=0}^{+\infty} \frac{1}{k!}(\mathbf{d} \cdot \nabla)^k \mathbf{f}(\mathbf{r}), \quad (0.8.13)$$

或者,

$$\mathbf{f}(\mathbf{r} + \mathbf{d}) = e^{\mathbf{d} \cdot \nabla} \mathbf{f}(\mathbf{r}). \quad (0.8.14)$$

§0.9 与位矢有关的微分

$$\nabla r = \frac{\mathbf{r}}{r} = \mathbf{e}_r, \quad r \neq 0. \quad (0.9.1)$$

$$\nabla \cdot \mathbf{r} = 3. \quad (0.9.2)$$

$$\nabla \times \mathbf{r} = 0. \quad (0.9.3)$$

$$\nabla(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}. \quad (0.9.4)$$

$$\nabla \psi(\varphi) = \frac{d\psi}{d\varphi} \nabla \varphi. \quad (0.9.5)$$

$$\nabla \psi(r) = \frac{d\psi}{dr} \frac{\mathbf{r}}{r}, \quad r \neq 0. \quad (0.9.6)$$

$$\nabla \frac{1}{r} = -\frac{\mathbf{r}}{r^3}, \quad r \neq 0. \quad (0.9.7)$$

$$\nabla \frac{1}{r^n} = -n \frac{\mathbf{r}}{r^{n+2}}, \quad r \neq 0, \quad n \in \mathbb{N}. \quad (0.9.8)$$

$$\nabla \cdot \frac{\mathbf{r}}{r^3} = 0, \quad r \neq 0. \quad (0.9.9)$$

$$\nabla \times \frac{\mathbf{r}}{r^3} = 0, \quad r \neq 0. \quad (0.9.10)$$

$$\nabla^2 \frac{1}{r} = 0, \quad r \neq 0. \quad (0.9.11)$$

$$\nabla \psi(\mathbf{r} - \mathbf{r}') = -\nabla' \psi(\mathbf{r} - \mathbf{r}'), \quad (0.9.12)$$

其中,

$$\nabla' = \mathbf{i} \frac{\partial}{\partial x'} + \mathbf{j} \frac{\partial}{\partial y'} + \mathbf{k} \frac{\partial}{\partial z'}. \quad (0.9.13)$$

$$\begin{aligned} \frac{1}{|\mathbf{r} - \mathbf{a}|} &= \frac{1}{r} + (-\mathbf{a} \cdot \nabla) \frac{1}{r} + \frac{1}{2} (-\mathbf{a} \cdot \nabla)^2 \frac{1}{r} + \cdots \\ &= \frac{1}{r} - \mathbf{a} \cdot \left(\nabla \frac{1}{r} \right) + \frac{1}{2} (-\mathbf{a} \cdot \nabla)^2 \frac{1}{r} + \cdots \\ &= \frac{1}{r} + (\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^3} + \frac{1}{2} (-\mathbf{a} \cdot \nabla) \left[(\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^3} \right] + \cdots \\ &= \frac{1}{r} + (\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^3} - \frac{1}{2} \mathbf{a} \cdot \left\{ \nabla \left[(\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^3} \right] \right\} + \cdots \end{aligned}$$

$$= \frac{1}{r} + (\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^3} + \frac{1}{2} \left[3(\mathbf{a} \cdot \mathbf{r})^2 \frac{1}{r^5} - \frac{\mathbf{a}^2}{r^3} \right] + \cdots \quad (0.9.14)$$

$$\begin{aligned} \frac{1}{|\mathbf{r} - \mathbf{a}|^3} &= \frac{1}{r^3} + (-\mathbf{a} \cdot \nabla) \frac{1}{r^3} + \cdots \\ &= \frac{1}{r^3} + 3(\mathbf{a} \cdot \mathbf{r}) \frac{1}{r^5} + \cdots . \end{aligned} \quad (0.9.15)$$

正交曲线坐标系下的场论公式

除直角坐标系外，柱坐标系和球坐标系也都是常用的坐标系。初看起来，柱坐标系和球坐标系似乎很复杂，但其实，它们同直角坐标系一样，都是正交归一右手系，只是坐标基底在空间上逐点变化而已。在直角坐标系下，坐标基底与空间上的点无关，不随空间点而变化，也就是说，在直角坐标系下，全空间可以共用一个坐标基底，这自然是最最简单的情况了。柱坐标系和球坐标系，虽然不很简单，但也并不复杂，这是因为它们都是正交曲线坐标系，属于一般曲线坐标系的特殊情形，因而较直角坐标系复杂，但又比一般的曲线坐标系简单。下面，我们先简单地介绍一下一般的曲线坐标系，然后讨论正交曲线坐标系下场论之计算，最后再将这些结果应用到柱坐标系和球坐标系上。

§0.10 曲线坐标系

§0.10.1 定义

在空间中任取一个直角坐标系，我们有

$$\mathbf{r} = x_1 \mathbf{i}_1 + x_2 \mathbf{i}_2 + x_3 \mathbf{i}_3, \quad (0.10.1)$$

其中， $\mathbf{r}$ 就是所谓的位矢，并且

$$x_1 = x, \quad x_2 = y, \quad x_3 = z, \quad (0.10.2)$$

$$\mathbf{i}_1 = \mathbf{i}, \quad \mathbf{i}_2 = \mathbf{j}, \quad \mathbf{i}_3 = \mathbf{k} \quad (0.10.3)$$

设有三个函数，

$$u_\alpha = f_\alpha(x_1, x_2, x_3), \quad \alpha = 1, 2, 3. \quad (0.10.4)$$

它们满足

$$J = \frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \frac{\partial u_1}{\partial x_3} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_2}{\partial x_3} \\ \frac{\partial u_3}{\partial x_1} & \frac{\partial u_3}{\partial x_2} & \frac{\partial u_3}{\partial x_3} \end{vmatrix} \neq 0. \quad (0.10.5)$$

由隐函数定理，我们得逆函数，

$$x_\alpha = g_\alpha(u_1, u_2, u_3), \quad \alpha = 1, 2, 3. \quad (0.10.6)$$

这说明下列的两个有序数组是一一对应的，

$$(x_1, x_2, x_3) \rightleftharpoons (u_1, u_2, u_3). \quad (0.10.7)$$

人们称 u_1, u_2, u_3 为参数系或曲线坐标系（曲紋坐标系），并称 (u_1, u_2, u_3) 为空间的曲线坐标（曲紋坐标）。

曲线坐标系有三个坐标曲面，

$$u_\alpha = c_\alpha, \quad \alpha = 1, 2, 3, \quad (0.10.8)$$

也就是，

$$f_\alpha(x_1, x_2, x_3) = c_\alpha, \quad \alpha = 1, 2, 3, \quad (0.10.9)$$

其中， c_α 是常数。亦有三个坐标曲线，即 u_1 曲线，

$$\begin{cases} f_2(x_1, x_2, x_3) = c_2 \\ f_3(x_1, x_2, x_3) = c_3, \end{cases} \quad (0.10.10)$$

u_2 曲线，

$$\begin{cases} f_1(x_1, x_2, x_3) = c_1 \\ f_3(x_1, x_2, x_3) = c_3, \end{cases} \quad (0.10.11)$$

和 u_3 曲线，

$$\begin{cases} f_1(x_1, x_2, x_3) = c_1 \\ f_2(x_1, x_2, x_3) = c_2. \end{cases} \quad (0.10.12)$$

显然，在 u_1 曲线上， u_2 和 u_3 固定， u_1 连续变化。同理，在 u_2 曲线上， u_1 和 u_3 固定， u_2 连续变化；在 u_3 曲线上， u_1 和 u_2 固定， u_3 连续变化。

在曲线坐标系下，

$$\mathbf{r} = x_1(u_1, u_2, u_3)\mathbf{i}_1 + x_2(u_1, u_2, u_3)\mathbf{i}_2 + x_3(u_1, u_2, u_3)\mathbf{i}_3. \quad (0.10.13)$$

可见，位矢 $\mathbf{r}$ 是曲线坐标的函数，

$$\mathbf{r} = \mathbf{r}(u_1, u_2, u_3). \quad (0.10.14)$$

这样，通过曲紋坐标，我们在空间任一点 (u_1, u_2, u_3) 可以引进三个矢量：

$$\frac{\partial \mathbf{r}}{\partial u_\alpha}, \quad \alpha = 1, 2, 3. \quad (0.10.15)$$

容易看出，它们分别是三个坐标曲线的切矢。这三个矢量是线性无关的，或者说，它们是不共面的，这是因为它们所生成的平行六面体的体积不等于零，

$$v = \begin{vmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \frac{\partial x_1}{\partial u_3} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \frac{\partial x_2}{\partial u_3} \\ \frac{\partial x_3}{\partial u_1} & \frac{\partial x_3}{\partial u_2} & \frac{\partial x_3}{\partial u_3} \end{vmatrix} = \frac{\partial(x_1, x_2, x_3)}{\partial(u_1, u_2, u_3)} = J^{-1} \neq 0. \quad (0.10.16)$$

因此，人们可以取这三个矢量作为点 (u_1, u_2, u_3) 处的基底。这样，空间上每一点都有一个属于自己的基底，逐点变化，各点基底一般各不相同。对于直角坐标系， $u_\alpha = x_\alpha$ ，易知

$$\frac{\partial \mathbf{r}}{\partial u_\alpha} = \mathbf{i}_\alpha, \quad \alpha = 1, 2, 3. \quad (0.10.17)$$

可见，直角坐标系是很特别的坐标系，它在空间每一点上所产生的基底都是一样的，从而不是逐点变化，而是固定不动的。

按照上面所说，我们在空间每一点都可以得到一个仿射标架或仿射坐标系，

$$\left\{ (u_1, u_2, u_3); \frac{\partial \mathbf{r}}{\partial u_1}, \frac{\partial \mathbf{r}}{\partial u_2}, \frac{\partial \mathbf{r}}{\partial u_3} \right\}, \quad (0.10.18)$$

人称**局部坐标系**。

上述局部坐标系的基矢来自于坐标曲线的切矢。代替这些切矢，我们也可以使用坐标曲面的法矢，它们分别是

$$\nabla u_\alpha = \mathbf{i}_1 \frac{\partial u_\alpha}{\partial x_1} + \mathbf{i}_2 \frac{\partial u_\alpha}{\partial x_2} + \mathbf{i}_3 \frac{\partial u_\alpha}{\partial x_3}, \quad \alpha = 1, 2, 3. \quad (0.10.19)$$

也就是，

$$\nabla f_\alpha = \mathbf{i}_1 \frac{\partial f_\alpha}{\partial x_1} + \mathbf{i}_2 \frac{\partial f_\alpha}{\partial x_2} + \mathbf{i}_3 \frac{\partial f_\alpha}{\partial x_3}, \quad \alpha = 1, 2, 3. \quad (0.10.20)$$

因为雅可比行列式 J 不等于零，所以以上的三个法矢也是线性无关的。这意味着

$$\{(u_1, u_2, u_3); \nabla u_1, \nabla u_2, \nabla u_3\}, \quad (0.10.21)$$

也是一个局部仿射标架或局部坐标系。

利用复合函数求导的链式法则，我们有

$$\sum_{\beta=1}^3 \frac{\partial u_\alpha}{\partial x_\beta} \frac{\partial x_\beta}{\partial u_\gamma} = \frac{\partial u_\alpha}{\partial u_\gamma} = \delta_{\alpha\gamma}. \quad (0.10.22)$$

这意味着

$$(\nabla u_\alpha) \cdot \frac{\partial \mathbf{r}}{\partial u_\beta} = \delta_{\alpha\beta}. \quad (0.10.23)$$

这说明，上述两个基底的基矢是相互垂直的 (bi-orthogonal)。象这样相互垂直的两个基底称为**倒易基底** (reciprocal base vectors)。相互倒易的两个基底，知其一，即可得其二。例如，因为

$$(\nabla u_1) \cdot \frac{\partial \mathbf{r}}{\partial u_2} = 0, \quad (\nabla u_1) \cdot \frac{\partial \mathbf{r}}{\partial u_3} = 0, \quad (0.10.24)$$

所以

$$\nabla u_1 \perp \frac{\partial \mathbf{r}}{\partial u_2}, \quad \nabla u_1 \perp \frac{\partial \mathbf{r}}{\partial u_3}. \quad (0.10.25)$$

于是，

$$\nabla u_1 \parallel \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right). \quad (0.10.26)$$

这说明

$$\nabla u_1 = \lambda \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right), \quad (0.10.27)$$

其中， λ 是某一实系数，待定。再因

$$(\nabla u_1) \cdot \frac{\partial \mathbf{r}}{\partial u_1} = 1, \quad (0.10.28)$$

所以

$$\lambda \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right) \cdot \frac{\partial \mathbf{r}}{\partial u_1} = 1. \quad (0.10.29)$$

解之, 得

$$\lambda = \frac{1}{\frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right)}. \quad (0.10.30)$$

这样, 我们便有

$$\nabla u_1 = \frac{\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3}}{\frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right)}. \quad (0.10.31)$$

同理,

$$\nabla u_2 = \frac{\frac{\partial \mathbf{r}}{\partial u_3} \times \frac{\partial \mathbf{r}}{\partial u_1}}{\frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right)}, \quad (0.10.32)$$

$$\nabla u_3 = \frac{\frac{\partial \mathbf{r}}{\partial u_1} \times \frac{\partial \mathbf{r}}{\partial u_2}}{\frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right)}. \quad (0.10.33)$$

反之, 易证

$$\frac{\partial \mathbf{r}}{\partial u_1} = \frac{\nabla u_2 \times \nabla u_3}{\nabla u_1 \cdot (\nabla u_2 \times \nabla u_3)}, \quad (0.10.34)$$

$$\frac{\partial \mathbf{r}}{\partial u_2} = \frac{\nabla u_3 \times \nabla u_1}{\nabla u_1 \cdot (\nabla u_2 \times \nabla u_3)}, \quad (0.10.35)$$

$$\frac{\partial \mathbf{r}}{\partial u_3} = \frac{\nabla u_1 \times \nabla u_2}{\nabla u_1 \cdot (\nabla u_2 \times \nabla u_3)}. \quad (0.10.36)$$

因为这两个基底有这样的关系, 所以任选其一即可, 通常选

$$\left\{ \frac{\partial \mathbf{r}}{\partial u_1}, \frac{\partial \mathbf{r}}{\partial u_2}, \frac{\partial \mathbf{r}}{\partial u_3} \right\} \quad (0.10.37)$$

作为正基底, 选

$$\left\{ (u_1, u_2, u_3); \frac{\partial \mathbf{r}}{\partial u_1}, \frac{\partial \mathbf{r}}{\partial u_2}, \frac{\partial \mathbf{r}}{\partial u_3} \right\} \quad (0.10.38)$$

作为局部坐标系。虽然如此, 知道倒基底是没有坏处的, 它也有很多的用处。

所谓正交曲线坐标系, 是指三个坐标基矢在空间每一点上均两两相互垂直,

$$\frac{\partial \mathbf{r}}{\partial u_1} \perp \frac{\partial \mathbf{r}}{\partial u_2}, \quad \frac{\partial \mathbf{r}}{\partial u_2} \perp \frac{\partial \mathbf{r}}{\partial u_3}, \quad \frac{\partial \mathbf{r}}{\partial u_3} \perp \frac{\partial \mathbf{r}}{\partial u_1}. \quad (0.10.39)$$

此时, 为了方便, 人们经常将上式三个矢量归一化,

$$\mathbf{e}_\alpha = \frac{1}{h_\alpha} \frac{\partial \mathbf{r}}{\partial u_\alpha}, \quad (0.10.40)$$

其中, h_α 是所谓的拉梅系数 (Lamé coefficient),

$$h_\alpha = \left| \frac{\partial \mathbf{r}}{\partial u_\alpha} \right|. \quad (0.10.41)$$

归一化不影响正交性, 因而, 我们有

$$\mathbf{e}_\alpha \cdot \mathbf{e}_\beta = \delta_{\alpha\beta}. \quad (0.10.42)$$

通常称这样的基底为正交归一基底。通过适当地调整三个参数 u_1, u_2, u_3 的次序, 我们还可以使得前述之基底成为右手系, 即

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \quad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \quad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2. \quad (0.10.43)$$

于是, 我们便在空间每一点上得到一个正交归一右手系。以下, 我们总是假定局部坐标系为正交归一右手系, 不再一一声明。

利用局部坐标系, 我们有

$$d\mathbf{r} = \sum_{\alpha=1}^3 \frac{\partial \mathbf{r}}{\partial u_\alpha} du_\alpha = \sum_{\alpha=1}^3 \mathbf{e}_\alpha h_\alpha du_\alpha, \quad (0.10.44)$$

其中,

$$h_\alpha = \left| \frac{\partial \mathbf{r}}{\partial u_\alpha} \right|, \quad \mathbf{e}_\alpha = \frac{1}{h_\alpha} \frac{\partial \mathbf{r}}{\partial u_\alpha}. \quad (0.10.45)$$

可见, 沿三个坐标曲线的弧元为

$$dl_\alpha = h_\alpha du_\alpha, \quad \alpha = 1, 2, 3. \quad (0.10.46)$$

沿空间任一曲线的弧元为

$$dl = \sqrt{d\mathbf{r} \cdot d\mathbf{r}} = \sqrt{\sum_{\alpha} h_\alpha^2 du_\alpha^2}. \quad (0.10.47)$$

三个坐标曲面上的面元为

$$ds_\alpha = h_\beta h_\gamma du_\beta du_\gamma, \quad \alpha, \beta, \gamma \text{按} 1, 2, 3 \text{循环}. \quad (0.10.48)$$

点 (u_1, u_2, u_3) 处的一般面元为

$$d\mathbf{s} = \sum_{\alpha} \mathbf{e}_\alpha ds_\alpha = \mathbf{e}_1 h_2 h_3 du_2 du_3 + \mathbf{e}_2 h_3 h_1 du_3 du_1 + \mathbf{e}_3 h_1 h_2 du_1 du_2. \quad (0.10.49)$$

点 (u_1, u_2, u_3) 处的体积元为

$$dv = \frac{\partial \mathbf{r}}{\partial u_1} du_1 \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} du_2 \times \frac{\partial \mathbf{r}}{\partial u_3} du_3 \right) = h_1 h_2 h_3 du_1 du_2 du_3. \quad (0.10.50)$$

这些都是常用的结果, 值得我们注意。

下面, 我们讨论正交曲线坐标系的两个常用例子。

§0.10.2 柱坐标系

柱坐标系的定义为

$$\begin{cases} x &= \rho \cos \varphi \\ y &= \rho \sin \varphi \\ z &= z, \end{cases} \quad (0.10.51)$$

其中,

$$0 \leq \rho < +\infty, \quad -\pi < \varphi \leq \pi, \quad -\infty < z < +\infty. \quad (0.10.52)$$

于是, 我们有

$$\frac{\partial \mathbf{r}}{\partial \rho} = \mathbf{i} \cos \varphi + \mathbf{j} \sin \varphi, \quad (0.10.53)$$

$$\frac{\partial \mathbf{r}}{\partial \varphi} = \mathbf{i}(-\rho \sin \varphi) + \mathbf{j} \rho \cos \varphi, \quad (0.10.54)$$

$$\frac{\partial \mathbf{r}}{\partial z} = \mathbf{k}. \quad (0.10.55)$$

容易证明这三个矢量是两两相互的, 即

$$\frac{\partial \mathbf{r}}{\partial \rho} \cdot \frac{\partial \mathbf{r}}{\partial \varphi} = 0, \quad \frac{\partial \mathbf{r}}{\partial \varphi} \cdot \frac{\partial \mathbf{r}}{\partial z} = 0, \quad \frac{\partial \mathbf{r}}{\partial z} \cdot \frac{\partial \mathbf{r}}{\partial \rho} = 0. \quad (0.10.56)$$

可见, 柱坐标系是正交曲线坐标系。这里, 三个拉梅系数分别为

$$h_\rho = \left| \frac{\partial \mathbf{r}}{\partial \rho} \right| = 1, \quad h_\varphi = \left| \frac{\partial \mathbf{r}}{\partial \varphi} \right| = \rho, \quad h_z = \left| \frac{\partial \mathbf{r}}{\partial z} \right| = 1. \quad (0.10.57)$$

于是, 归一化, 得

$$\mathbf{e}_\rho = \frac{1}{h_\rho} \frac{\partial \mathbf{r}}{\partial \rho} = \mathbf{i} \cos \varphi + \mathbf{j} \sin \varphi, \quad (0.10.58)$$

$$\mathbf{e}_\varphi = \frac{1}{h_\varphi} \frac{\partial \mathbf{r}}{\partial \varphi} = \mathbf{i}(-\sin \varphi) + \mathbf{j} \cos \varphi, \quad (0.10.59)$$

$$\mathbf{e}_z = \frac{1}{h_z} \frac{\partial \mathbf{r}}{\partial z} = \mathbf{k}. \quad (0.10.60)$$

这样, 我们便有

$$\mathbf{e}_\alpha \cdot \mathbf{e}_\beta = \delta_{\alpha\beta}, \quad \alpha, \beta = \rho, \varphi, z. \quad (0.10.61)$$

如果约定三个曲线坐标的次序为 ρ, φ, z , 则容易验证 $\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{e}_z$ 为右手系, 即

$$\mathbf{e}_\rho \times \mathbf{e}_\varphi = \mathbf{e}_z, \quad \mathbf{e}_\varphi \times \mathbf{e}_z = \mathbf{e}_\rho, \quad \mathbf{e}_z \times \mathbf{e}_\rho = \mathbf{e}_\varphi. \quad (0.10.62)$$

总而言之, 柱坐标系的局部基底为 $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{e}_z\}$, 它是一个正交归一右手系。

在柱坐标系下, 有

$$d\mathbf{r} = \mathbf{e}_\rho d\rho + \mathbf{e}_\varphi \rho d\varphi + \mathbf{e}_z dz, \quad (0.10.63)$$

$$ds = \mathbf{e}_\rho \rho d\varphi dz + \mathbf{e}_\varphi d\rho dz + \mathbf{e}_z \rho d\rho d\varphi, \quad (0.10.64)$$

$$dv = \rho d\rho d\varphi dz. \quad (0.10.65)$$

§0.10.3 球坐标系

球坐标系的定义为

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta, \end{cases} \quad (0.10.66)$$

其中,

$$0 \leq r < +\infty, \quad 0 \leq \theta < \pi \quad -\pi < \varphi \leq \pi. \quad (0.10.67)$$

我们有

$$\frac{\partial \mathbf{r}}{\partial r} = \mathbf{i} \sin \theta \cos \varphi + \mathbf{j} \sin \theta \sin \varphi + \mathbf{k} \cos \theta, \quad (0.10.68)$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = \mathbf{i} r \cos \theta \cos \varphi + \mathbf{j} r \cos \theta \sin \varphi + \mathbf{k}(-r \sin \theta), \quad (0.10.69)$$

$$\frac{\partial \mathbf{r}}{\partial \varphi} = \mathbf{i}(-r \sin \theta \sin \varphi) + \mathbf{j} r \sin \theta \cos \varphi. \quad (0.10.70)$$

这三个矢量也是两两相互正交的,

$$\frac{\partial \mathbf{r}}{\partial r} \cdot \frac{\partial \mathbf{r}}{\partial \theta} = 0, \quad \frac{\partial \mathbf{r}}{\partial \theta} \cdot \frac{\partial \mathbf{r}}{\partial \varphi} = 0, \quad \frac{\partial \mathbf{r}}{\partial r} \cdot \frac{\partial \mathbf{r}}{\partial \varphi} = 0. \quad (0.10.71)$$

可见, 同柱坐标系一样, 球坐标系也是正交曲线坐标系。它的三个拉梅系数分别为

$$h_r = \left| \frac{\partial \mathbf{r}}{\partial r} \right| = 1, \quad h_\theta = \left| \frac{\partial \mathbf{r}}{\partial \theta} \right| = r, \quad h_\varphi = \left| \frac{\partial \mathbf{r}}{\partial \varphi} \right| = r \sin \theta. \quad (0.10.72)$$

归一化, 得

$$\mathbf{e}_r = \frac{1}{h_r} \frac{\partial \mathbf{r}}{\partial r} = \mathbf{i} \sin \theta \cos \varphi + \mathbf{j} \sin \theta \sin \varphi + \mathbf{k} \cos \theta, \quad (0.10.73)$$

$$\mathbf{e}_\theta = \frac{1}{h_\theta} \frac{\partial \mathbf{r}}{\partial \theta} = \mathbf{i} \cos \theta \cos \varphi + \mathbf{j} \cos \theta \sin \varphi + \mathbf{k}(-\sin \theta), \quad (0.10.74)$$

$$\mathbf{e}_\varphi = \frac{1}{h_\varphi} \frac{\partial \mathbf{r}}{\partial \varphi} = \mathbf{i}(-\sin \varphi) + \mathbf{j} \cos \varphi. \quad (0.10.75)$$

于是,

$$\mathbf{e}_\alpha \cdot \mathbf{e}_\beta = \delta_{\alpha\beta}, \quad \alpha, \beta = r, \theta, \varphi. \quad (0.10.76)$$

现在, 让我们约定三个曲线坐标的次序为 r, θ, φ 。容易验证

$$\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{e}_\varphi, \quad \mathbf{e}_\theta \times \mathbf{e}_\varphi = \mathbf{e}_r, \quad \mathbf{e}_\varphi \times \mathbf{e}_r = \mathbf{e}_\theta. \quad (0.10.77)$$

这样, $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$ 就构成了一个正交归一右手系, 它是球坐标系的局部基底。

在球坐标系下, 我们有

$$d\mathbf{r} = \mathbf{e}_r dr + \mathbf{e}_\theta r d\theta + \mathbf{e}_\varphi r \sin \theta d\varphi, \quad (0.10.78)$$

$$ds = \mathbf{e}_r r^2 \sin \theta d\theta d\varphi + \mathbf{e}_\theta r \sin \theta dr d\varphi + \mathbf{e}_\varphi r dr d\theta, \quad (0.10.79)$$

$$dv = r^2 \sin \theta dr d\theta d\varphi. \quad (0.10.80)$$

§0.11 正交曲线坐标系下场论之计算

§0.11.1 梯度

设 ψ 为标量场,

$$\psi = \psi(u_1, u_2, u_3). \quad (0.11.1)$$

令

$$\nabla\psi = \lambda_1\mathbf{e}_1 + \lambda_2\mathbf{e}_2 + \lambda_3\mathbf{e}_3. \quad (0.11.2)$$

这里, λ_1 其实就是标量场 ψ 沿 $\mathbf{e}_1$ 的方向导数, λ_2, λ_3 的意义是类似的。考虑到

$$d\psi = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy + \frac{\partial\psi}{\partial z}dz = (\nabla\psi) \cdot d\mathbf{r}, \quad (0.11.3)$$

以及

$$d\psi = \frac{\partial\psi}{\partial u_1}du_1 + \frac{\partial\psi}{\partial u_2}du_2 + \frac{\partial\psi}{\partial u_3}du_3, \quad (0.11.4)$$

$$d\mathbf{r} = \mathbf{e}_1 h_1 du_1 + \mathbf{e}_2 h_2 du_2 + \mathbf{e}_3 h_3 du_3, \quad (0.11.5)$$

我们有

$$\frac{\partial\psi}{\partial u_1}du_1 + \frac{\partial\psi}{\partial u_2}du_2 + \frac{\partial\psi}{\partial u_3}du_3 = \lambda_1 h_1 du_1 + \lambda_2 h_2 du_2 + \lambda_3 h_3 du_3, \quad (0.11.6)$$

或者,

$$\left(\frac{\partial\psi}{\partial u_1} - \lambda_1 h_1\right)du_1 + \left(\frac{\partial\psi}{\partial u_2} - \lambda_2 h_2\right)du_2 + \left(\frac{\partial\psi}{\partial u_3} - \lambda_3 h_3\right)du_3 = 0. \quad (0.11.7)$$

注意到

$$du_1 = \Delta u_1, \quad du_2 = \Delta u_2, \quad du_3 = \Delta u_3, \quad (0.11.8)$$

因此, du_1, du_2, du_3 是相互独立的变量。换句话说, 它们是线性无关的。是故, 我们有

$$\frac{\partial\psi}{\partial u_1} - \lambda_1 h_1 = 0, \quad \frac{\partial\psi}{\partial u_2} - \lambda_2 h_2 = 0, \quad \frac{\partial\psi}{\partial u_3} - \lambda_3 h_3 = 0. \quad (0.11.9)$$

解之, 得

$$\lambda_1 = \frac{1}{h_1} \frac{\partial\psi}{\partial u_1}, \quad \lambda_2 = \frac{1}{h_2} \frac{\partial\psi}{\partial u_2}, \quad \lambda_3 = \frac{1}{h_3} \frac{\partial\psi}{\partial u_3}. \quad (0.11.10)$$

将之代入式(0.11.2), 得

$$\nabla\psi = \frac{1}{h_1} \frac{\partial\psi}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial\psi}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial\psi}{\partial u_3} \mathbf{e}_3. \quad (0.11.11)$$

这就是正交曲线坐标系下标量场梯度的表达式。

§0.11.2 散度

设 $\mathbf{f}$ 为矢量场,

$$\mathbf{f} = f_1(u_1, u_2, u_3)\mathbf{e}_1 + f_2(u_1, u_2, u_3)\mathbf{e}_2 + f_3(u_1, u_2, u_3)\mathbf{e}_3. \quad (0.11.12)$$

考虑到

$$u_1 = u_1(x, y, z), \quad (0.11.13)$$

因此, u_1 乃空间上的标量场。既然 u_1 为标量场, 对之取梯度, 并利用上小节公式(0.11.11), 得

$$\nabla u_1 = \frac{1}{h_1} \mathbf{e}_1. \quad (0.11.14)$$

容易看出, ∇u_1 其实就是坐标曲面,

$$u_1(x, y, z) = c, \quad (0.11.15)$$

的法矢, 也就是倒基矢。下面将会看到, 倒基矢大有用处。

同理,

$$\nabla u_2 = \frac{1}{h_2} \mathbf{e}_2, \quad \nabla u_3 = \frac{1}{h_3} \mathbf{e}_3. \quad (0.11.16)$$

由是, 得

$$\mathbf{e}_1 = h_1 \nabla u_1, \quad \mathbf{e}_2 = h_2 \nabla u_2, \quad \mathbf{e}_3 = h_3 \nabla u_3. \quad (0.11.17)$$

利用正倒基底的倒易关系, 也能得到上述这些关系。

由以上结果, 我们有

$$\begin{aligned} \mathbf{f} &= f_1 \mathbf{e}_1 + f_2 \mathbf{e}_2 + f_3 \mathbf{e}_3 \\ &= f_1 \mathbf{e}_2 \times \mathbf{e}_3 + f_2 \mathbf{e}_3 \times \mathbf{e}_1 + f_3 \mathbf{e}_1 \times \mathbf{e}_2 \\ &= f_1 h_2 h_3 \nabla u_2 \times \nabla u_3 + f_2 h_3 h_1 \nabla u_3 \times \nabla u_1 + f_3 h_1 h_2 \nabla u_1 \times \nabla u_2. \end{aligned} \quad (0.11.18)$$

取散度, 得

$$\nabla \cdot \mathbf{f} = \nabla \cdot (f_1 h_2 h_3 \nabla u_2 \times \nabla u_3) + \nabla \cdot (f_2 h_3 h_1 \nabla u_3 \times \nabla u_1) + \nabla \cdot (f_3 h_1 h_2 \nabla u_1 \times \nabla u_2). \quad (0.11.19)$$

注意到

$$\nabla \cdot (\psi \mathbf{f}) = (\nabla \psi) \cdot \mathbf{f} + \psi \nabla \cdot \mathbf{f}, \quad (0.11.20)$$

我们有

$$\begin{aligned} &\nabla \cdot (f_1 h_2 h_3 \nabla u_2 \times \nabla u_3) \\ &= \nabla(f_1 h_2 h_3) \cdot (\nabla u_2 \times \nabla u_3) + f_1 h_2 h_3 \nabla \cdot (\nabla u_2 \times \nabla u_3). \end{aligned} \quad (0.11.21)$$

右边第一项为

$$\begin{aligned} &\nabla(f_1 h_2 h_3) \cdot (\nabla u_2 \times \nabla u_3) \\ &= \nabla(f_1 h_2 h_3) \cdot \left(\frac{\mathbf{e}_2}{h_2} \times \frac{\mathbf{e}_3}{h_3} \right) \\ &= \frac{1}{h_2 h_3} \nabla(f_1 h_2 h_3) \cdot \mathbf{e}_1 \\ &= \frac{1}{h_2 h_3} \left[\frac{1}{h_1} \frac{\partial(f_1 h_2 h_3)}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial(f_1 h_2 h_3)}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial(f_1 h_2 h_3)}{\partial u_3} \mathbf{e}_3 \right] \cdot \mathbf{e}_1 \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial(f_1 h_2 h_3)}{\partial u_1}. \end{aligned} \quad (0.11.22)$$

又,

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = (\nabla \times \mathbf{f}) \cdot \mathbf{g} - (\nabla \times \mathbf{g}) \cdot \mathbf{f}, \quad (0.11.23)$$

我们有

$$\begin{aligned}\nabla \cdot (\nabla u_2 \times \nabla u_3) &= [\nabla \times (\nabla u_2)] \cdot (\nabla u_3) - [\nabla \times (\nabla u_3)] \cdot (\nabla u_2) \\ &= 0,\end{aligned}\quad (0.11.24)$$

这是因为梯度的旋度恒为零。将以上结果代入式(0.11.21), 得

$$\nabla \cdot (f_1 h_2 h_3 \nabla u_2 \times \nabla u_3) = \frac{1}{h_1 h_2 h_3} \frac{\partial(f_1 h_2 h_3)}{\partial u_1}. \quad (0.11.25)$$

同理, 得

$$\nabla \cdot (f_2 h_3 h_1 \nabla u_3 \times \nabla u_1) = \frac{1}{h_1 h_2 h_3} \frac{\partial(f_2 h_3 h_1)}{\partial u_2} \quad (0.11.26)$$

$$\nabla \cdot (f_3 h_1 h_2 \nabla u_3 \times \nabla u_1) = \frac{1}{h_1 h_2 h_3} \frac{\partial(f_3 h_1 h_2)}{\partial u_3}. \quad (0.11.27)$$

将它们返回式(0.11.19), 我们得

$$\nabla \cdot \mathbf{f} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_1 h_2 h_3)}{\partial u_1} + \frac{\partial(f_2 h_3 h_1)}{\partial u_2} + \frac{\partial(f_3 h_1 h_2)}{\partial u_3} \right]. \quad (0.11.28)$$

这就是正交曲线坐标系下矢量场散度的计算公式。

我们也可以用对高斯公式求体积导数的方式来计算矢量场的散度。以点 (u_1, u_2, u_3) 为中心, 用三线元,

$$d\mathbf{l}_1 = h_1 du_1 \mathbf{e}_1, \quad d\mathbf{l}_2 = h_2 du_2 \mathbf{e}_2, \quad d\mathbf{l}_3 = h_3 du_3 \mathbf{e}_3, \quad (0.11.29)$$

张成一个小长方体。在此小长方体上应用高斯公式,

$$\oint \mathbf{f} \cdot d\mathbf{s} = \int \nabla \cdot \mathbf{f} dv, \quad (0.11.30)$$

得

$$\begin{aligned}& \left[-f_1 h_2 h_3 \Big|_{u_1 - du_1/2} du_2 du_3 + f_1 h_2 h_3 \Big|_{u_1 + du_1/2} du_2 du_3 \right] \\ & + \left[-f_2 h_3 h_1 \Big|_{u_2 - du_2/2} du_3 du_1 + f_2 h_3 h_1 \Big|_{u_2 + du_2/2} du_3 du_1 \right] \\ & + \left[-f_3 h_1 h_2 \Big|_{u_3 - du_3/2} du_1 du_2 + f_3 h_1 h_2 \Big|_{u_3 + du_3/2} du_1 du_2 \right] \\ & = (\nabla \cdot \mathbf{f}) h_1 h_2 h_3 du_1 du_2 du_3 + o(du_1 du_2 du_3).\end{aligned}\quad (0.11.31)$$

上式左边是长方体三对面上流进、流出的通量, 右边是散度乘以长方体的体积, 二者最多相差一个比体积高阶的小量。考虑到

$$-f_1 h_2 h_3 \Big|_{u_1 - du_1/2} + f_1 h_2 h_3 \Big|_{u_1 + du_1/2} = \frac{\partial(f_1 h_2 h_3)}{\partial u_1} du_1 + o(du_1), \quad (0.11.32)$$

$$-f_2 h_3 h_1 \Big|_{u_2 - du_2/2} + f_2 h_3 h_1 \Big|_{u_2 + du_2/2} = \frac{\partial(f_2 h_3 h_1)}{\partial u_2} du_2 + o(du_2), \quad (0.11.33)$$

$$-f_3 h_1 h_2 \Big|_{u_3 - du_3/2} + f_3 h_1 h_2 \Big|_{u_3 + du_3/2} = \frac{\partial(f_3 h_1 h_2)}{\partial u_3} du_3 + o(du_3), \quad (0.11.34)$$

我们有

$$\begin{aligned} & \left[\frac{\partial(f_1 h_2 h_3)}{\partial u_1} + \frac{\partial(f_2 h_3 h_1)}{\partial u_2} + \frac{\partial(f_3 h_1 h_2)}{\partial u_3} \right] du_1 du_2 du_3 \\ &= (\nabla \cdot \mathbf{f}) h_1 h_2 h_3 du_1 du_2 du_3 + o(du_1 du_2 du_3), \end{aligned} \quad (0.11.35)$$

两边除以小长方体的体积 $h_1 h_2 h_3 du_1 du_2 du_3$ ，并取极限（所谓体积导数是也）：

$$\Delta u_1 \rightarrow 0, \quad \Delta u_2 \rightarrow 0, \quad \Delta u_3 \rightarrow 0, \quad (0.11.36)$$

得

$$\nabla \cdot \mathbf{f} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_1 h_2 h_3)}{\partial u_1} + \frac{\partial(f_2 h_3 h_1)}{\partial u_2} + \frac{\partial(f_3 h_1 h_2)}{\partial u_3} \right]. \quad (0.11.37)$$

这样，我们就再次得到了正交曲线坐标系下矢量场散度的计算公式。这个推导让我们可以清楚看到，诸如 $f_1 h_2 h_3$ 以及 $h_1 h_2 h_3$ 之类的量是怎么出现的。

§0.11.3 旋度

同前，设 $\mathbf{f}$ 为矢量场，

$$\mathbf{f} = f_1(u_1, u_2, u_3)\mathbf{e}_1 + f_2(u_1, u_2, u_3)\mathbf{e}_2 + f_3(u_1, u_2, u_3)\mathbf{e}_3. \quad (0.11.38)$$

利用式(0.11.17)，得

$$\mathbf{f} = f_1 h_1 \nabla u_1 + f_2 h_2 \nabla u_2 + f_3 h_3 \nabla u_3. \quad (0.11.39)$$

取旋度，得

$$\nabla \times \mathbf{f} = \nabla \times (f_1 h_1 \nabla u_1) + \nabla \times (f_2 h_2 \nabla u_2) + \nabla \times (f_3 h_3 \nabla u_3). \quad (0.11.40)$$

注意到

$$\nabla \times (\psi \mathbf{f}) = (\nabla \psi) \times \mathbf{f} + \psi \nabla \times \mathbf{f}, \quad (0.11.41)$$

我们有

$$\nabla \times (f_1 h_1 \nabla u_1) = [\nabla(f_1 h_1)] \times \nabla u_1 + f_1 h_1 \nabla \times (\nabla u_1). \quad (0.11.42)$$

由于梯度的旋度恒为零，上式右边的第二项等于零。至于第一项，利用式(0.11.17)，得

$$\begin{aligned} [\nabla(f_1 h_1)] \times \nabla u_1 &= \frac{1}{h_1} [\nabla(f_1 h_1)] \times \mathbf{e}_1 \\ &= \frac{1}{h_1} \left[\frac{1}{h_1} \frac{\partial(f_1 h_1)}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial(f_1 h_1)}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial(f_1 h_1)}{\partial u_3} \mathbf{e}_3 \right] \times \mathbf{e}_1 \\ &= \frac{1}{h_3 h_1} \frac{\partial(f_1 h_1)}{\partial u_3} \mathbf{e}_2 - \frac{1}{h_1 h_2} \frac{\partial(f_1 h_1)}{\partial u_2} \mathbf{e}_3. \end{aligned} \quad (0.11.43)$$

这样，我们得

$$\nabla \times (f_1 h_1 \nabla u_1) = \frac{1}{h_3 h_1} \frac{\partial(f_1 h_1)}{\partial u_3} \mathbf{e}_2 - \frac{1}{h_1 h_2} \frac{\partial(f_1 h_1)}{\partial u_2} \mathbf{e}_3. \quad (0.11.44)$$

同理可得,

$$\nabla \times (f_2 h_2 \nabla u_2) = \frac{1}{h_1 h_2} \frac{\partial(f_2 h_2)}{\partial u_1} \mathbf{e}_3 - \frac{1}{h_2 h_3} \frac{\partial(f_2 h_2)}{\partial u_3} \mathbf{e}_1, \quad (0.11.45)$$

$$\nabla \times (f_3 h_3 \nabla u_3) = \frac{1}{h_2 h_3} \frac{\partial(f_3 h_3)}{\partial u_2} \mathbf{e}_1 - \frac{1}{h_3 h_1} \frac{\partial(f_3 h_3)}{\partial u_1} \mathbf{e}_2. \quad (0.11.46)$$

将以上结果代入式(0.11.40), 得

$$\begin{aligned} \nabla \times \mathbf{f} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_3 h_3)}{\partial u_2} - \frac{\partial(f_2 h_2)}{\partial u_3} \right] (h_1 \mathbf{e}_1) \\ &\quad + \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_1 h_1)}{\partial u_3} - \frac{\partial(f_3 h_3)}{\partial u_1} \right] (h_2 \mathbf{e}_2) \\ &\quad + \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_2 h_2)}{\partial u_1} - \frac{\partial(f_1 h_1)}{\partial u_2} \right] (h_3 \mathbf{e}_3). \end{aligned} \quad (0.11.47)$$

上式也可以用行列式表为如下形式,

$$\nabla \times \mathbf{f} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ f_1 h_1 & f_2 h_2 & f_3 h_3 \end{vmatrix}. \quad (0.11.48)$$

以上两式就是正交曲线坐标系下矢量场旋度的计算公式。

我们也可以通过Stokes公式求面积导数的方式来计算矢量场的旋度。设

$$\nabla \times \mathbf{f} = \lambda_1 \mathbf{e}_1 + \lambda_2 \mathbf{e}_2 + \lambda_3 \mathbf{e}_3, \quad (0.11.49)$$

其中, $\lambda_1, \lambda_2, \lambda_3$ 是矢量场 $\mathbf{f}$ 沿 $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ 的三个方向旋量。下面, 我们分别来计算 $\lambda_1, \lambda_2, \lambda_3$, 从 λ_1 开始。

以点 (u_1, u_2, u_3) 为中心, 用以下的两个线元,

$$d\mathbf{l}_2 = h_2 du_2 \mathbf{e}_2, \quad d\mathbf{l}_3 = h_3 du_3 \mathbf{e}_3, \quad (0.11.50)$$

张成一个小长方形, 此长方形的正向平行于 $\mathbf{e}_1$ 。现在, 将Stokes公式应用到此小长方形上,

$$\oint \mathbf{f} \cdot d\mathbf{l} = \int \nabla \times \mathbf{f} \cdot d\mathbf{s}, \quad (0.11.51)$$

我们有

$$\begin{aligned} &\left[f_2 h_2 \Big|_{u_3 - du_3/2} du_2 - f_2 h_2 \Big|_{u_3 + du_3/2} du_2 \right] + \left[f_3 h_3 \Big|_{u_2 + du_2/2} du_3 - f_3 h_3 \Big|_{u_2 - du_2/2} du_3 \right] \\ &= \lambda_1 h_2 h_3 du_1 du_2 + o(du_1 du_2). \end{aligned} \quad (0.11.52)$$

上式的左边是沿长方形四边的环流, 右边是方向旋量乘以长方形的面积, 二者最多相差一个比面积高阶的小量。由于

$$f_2 h_2 \Big|_{u_3 - du_3/2} - f_2 h_2 \Big|_{u_3 + du_3/2} = -\frac{\partial(f_2 h_2)}{\partial u_3} du_3 + o(du_3), \quad (0.11.53)$$

$$f_3 h_3 \Big|_{u_2 + du_2/2} - f_3 h_3 \Big|_{u_2 - du_2/2} = \frac{\partial(f_3 h_3)}{\partial u_2} du_2 + o(du_2), \quad (0.11.54)$$

因此,

$$\left[-\frac{\partial(f_2 h_2)}{\partial u_3} + \frac{\partial(f_3 h_3)}{\partial u_2} \right] du_2 du_3 = \lambda_1 h_2 h_3 du_1 du_2 + o(du_1 du_2), \quad (0.11.55)$$

两边除以小长方形的面积 $h_2 h_3 du_2 du_3$, 并取极限 (所谓面积导数是也):

$$\Delta u_2 \rightarrow 0, \quad \Delta u_3 \rightarrow 0, \quad (0.11.56)$$

得

$$\lambda_1 = \frac{1}{h_2 h_3} \left[\frac{\partial(f_3 h_3)}{\partial u_2} - \frac{\partial(f_2 h_2)}{\partial u_3} \right]. \quad (0.11.57)$$

同理可得

$$\lambda_2 = \frac{1}{h_3 h_1} \left[\frac{\partial(f_1 h_1)}{\partial u_3} - \frac{\partial(f_3 h_3)}{\partial u_1} \right], \quad (0.11.58)$$

$$\lambda_3 = \frac{1}{h_1 h_2} \left[\frac{\partial(f_2 h_2)}{\partial u_1} - \frac{\partial(f_1 h_1)}{\partial u_2} \right]. \quad (0.11.59)$$

将它们代入式(0.11.49), 得

$$\begin{aligned} \nabla \times \mathbf{f} &= \frac{1}{h_2 h_3} \left[\frac{\partial(f_3 h_3)}{\partial u_2} - \frac{\partial(f_2 h_2)}{\partial u_3} \right] \mathbf{e}_1 + \frac{1}{h_1 h_3} \left[\frac{\partial(f_1 h_1)}{\partial u_3} - \frac{\partial(f_3 h_3)}{\partial u_1} \right] \mathbf{e}_2 \\ &\quad + \frac{1}{h_1 h_2} \left[\frac{\partial(f_2 h_2)}{\partial u_1} - \frac{\partial(f_1 h_1)}{\partial u_2} \right] \mathbf{e}_3. \end{aligned} \quad (0.11.60)$$

此与式(0.11.47)一致, 我们再次得到了正交曲线坐标系下矢量场旋度的计算公式。这个推导显示了 $f_1 h_1$ 以及 $h_1 h_2$ 诸量的来源。

§0.11.4 拉普拉斯算子

同前, 设 ψ 为标量场,

$$\psi = \psi(u_1, u_2, u_3). \quad (0.11.61)$$

考虑到

$$\nabla^2 \psi = \nabla \cdot (\nabla \psi), \quad (0.11.62)$$

我们有

$$\begin{aligned} \nabla^2 \psi &= \nabla \cdot \left(\frac{1}{h_1} \frac{\partial \psi}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial \psi}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial \psi}{\partial u_3} \mathbf{e}_3 \right) \\ &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial u_3} \right) \right]. \end{aligned} \quad (0.11.63)$$

这样, 我们有

$$\nabla^2 \psi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial u_3} \right) \right], \quad (0.11.64)$$

或者

$$\nabla^2 \psi = \frac{1}{h_1 h_2 h_3} \sum_{\alpha=1}^3 \frac{\partial}{\partial u_\alpha} \left(\frac{h_1 h_2 h_3}{h_\alpha^2} \frac{\partial \psi}{\partial u_\alpha} \right). \quad (0.11.65)$$

以上就是正交曲线坐标系下拉普拉斯算子的计算公式。

§0.12 对柱坐标系的应用

对柱坐标系, 我们有

$$u_1 = \rho, \quad u_2 = \varphi, \quad u_3 = z, \quad (0.12.1)$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\varphi = \rho, \quad h_3 = h_z = 1, \quad (0.12.2)$$

$$\mathbf{e}_1 = \mathbf{e}_\rho, \quad \mathbf{e}_2 = \mathbf{e}_\varphi, \quad \mathbf{e}_3 = \mathbf{e}_z. \quad (0.12.3)$$

设 ψ 为标量场, $\mathbf{f}$ 为矢量场, 则

$$\psi = \psi(\rho, \varphi, z), \quad (0.12.4)$$

$$\mathbf{f} = f_\rho(\rho, \varphi, z)\mathbf{e}_\rho + f_\varphi(\rho, \varphi, z)\mathbf{e}_\varphi + f_z(\rho, \varphi, z)\mathbf{e}_z. \quad (0.12.5)$$

标量场的梯度为

$$\begin{aligned} \nabla\psi &= \frac{1}{h_1} \frac{\partial\psi}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial\psi}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial\psi}{\partial u_3} \mathbf{e}_3 \\ &= \frac{\partial\psi}{\partial\rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial\psi}{\partial\varphi} \mathbf{e}_\varphi + \frac{\partial\psi}{\partial z} \mathbf{e}_z. \end{aligned} \quad (0.12.6)$$

矢量场的散度为

$$\begin{aligned} \nabla \cdot \mathbf{f} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_1 h_2 h_3)}{\partial u_1} + \frac{\partial(f_2 h_3 h_1)}{\partial u_2} + \frac{\partial(f_3 h_1 h_2)}{\partial u_3} \right] \\ &= \frac{1}{\rho} \left[\frac{\partial(f_\rho \rho)}{\partial \rho} + \frac{\partial f_\varphi}{\partial \varphi} + \frac{\partial(f_z \rho)}{\partial z} \right] \\ &= \frac{1}{\rho} \frac{\partial(\rho f_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial f_\varphi}{\partial \varphi} + \frac{\partial f_z}{\partial z}. \end{aligned} \quad (0.12.7)$$

矢量场的旋度为

$$\begin{aligned} \nabla \times \mathbf{f} &= \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ f_1 h_1 & f_2 h_2 & f_3 h_3 \end{vmatrix} \\ &= \frac{1}{\rho} \begin{vmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\varphi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ f_\rho & f_\varphi \rho & f_z \end{vmatrix} \\ &= \left(\frac{1}{\rho} \frac{\partial f_z}{\partial \varphi} - \frac{\partial f_\varphi}{\partial z} \right) \mathbf{e}_\rho + \left(\frac{\partial f_\rho}{\partial z} - \frac{\partial f_z}{\partial \rho} \right) \mathbf{e}_\varphi + \frac{1}{\rho} \left(\frac{\partial(\rho f_\varphi)}{\partial \rho} - \frac{\partial f_\rho}{\partial \varphi} \right) \mathbf{e}_z. \end{aligned} \quad (0.12.8)$$

拉普拉斯算子为

$$\begin{aligned} \nabla^2 \psi &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial u_3} \right) \right] \\ &= \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{\partial}{\partial \varphi} \left(\frac{1}{\rho} \frac{\partial \psi}{\partial \varphi} \right) + \frac{\partial}{\partial z} \left(\rho \frac{\partial \psi}{\partial z} \right) \right] \\ &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \varphi^2} + \frac{\partial^2 \psi}{\partial z^2}. \end{aligned} \quad (0.12.9)$$

总而言之，在柱坐标系下，我们有

$$\nabla\psi = \frac{\partial\psi}{\partial\rho}\mathbf{e}_\rho + \frac{1}{\rho}\frac{\partial\psi}{\partial\varphi}\mathbf{e}_\varphi + \frac{\partial\psi}{\partial z}\mathbf{e}_z. \quad (0.12.10)$$

$$\nabla \cdot \mathbf{f} = \frac{1}{\rho}\frac{\partial(\rho f_\rho)}{\partial\rho} + \frac{1}{\rho}\frac{\partial f_\varphi}{\partial\varphi} + \frac{\partial f_z}{\partial z}. \quad (0.12.11)$$

$$\nabla \times \mathbf{f} = \left(\frac{1}{\rho}\frac{\partial f_z}{\partial\varphi} - \frac{\partial f_\varphi}{\partial z}\right)\mathbf{e}_\rho + \left(\frac{\partial f_\rho}{\partial z} - \frac{\partial f_z}{\partial\rho}\right)\mathbf{e}_\varphi + \frac{1}{\rho}\left(\frac{\partial(\rho f_\varphi)}{\partial\rho} - \frac{\partial f_\rho}{\partial\varphi}\right)\mathbf{e}_z. \quad (0.12.12)$$

$$\nabla^2\psi = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\varphi^2} + \frac{\partial^2\psi}{\partial z^2}. \quad (0.12.13)$$

§0.13 对球坐标系的应用

对球坐标系，我们有

$$u_1 = r, \quad u_2 = \theta, \quad u_3 = \varphi, \quad (0.13.1)$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_\varphi = r \sin \theta, \quad (0.13.2)$$

$$\mathbf{e}_1 = \mathbf{e}_r, \quad \mathbf{e}_2 = \mathbf{e}_\theta, \quad \mathbf{e}_3 = \mathbf{e}_\varphi. \quad (0.13.3)$$

同前，设 ψ 为标量场， $\mathbf{f}$ 为矢量场，则

$$\psi = \psi(r, \theta, \varphi), \quad (0.13.4)$$

$$\mathbf{f} = f_r(r, \theta, \varphi)\mathbf{e}_r + f_\theta(r, \theta, \varphi)\mathbf{e}_\theta + f_\varphi(r, \theta, \varphi)\mathbf{e}_\varphi. \quad (0.13.5)$$

标量场的梯度为

$$\begin{aligned} \nabla\psi &= \frac{1}{h_1}\frac{\partial\psi}{\partial u_1}\mathbf{e}_1 + \frac{1}{h_2}\frac{\partial\psi}{\partial u_2}\mathbf{e}_2 + \frac{1}{h_3}\frac{\partial\psi}{\partial u_3}\mathbf{e}_3 \\ &= \frac{\partial\psi}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\mathbf{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\varphi}\mathbf{e}_\varphi. \end{aligned} \quad (0.13.6)$$

矢量场的散度为

$$\begin{aligned} \nabla \cdot \mathbf{f} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(f_1 h_2 h_3)}{\partial u_1} + \frac{\partial(f_2 h_3 h_1)}{\partial u_2} + \frac{\partial(f_3 h_1 h_2)}{\partial u_3} \right] \\ &= \frac{1}{r^2 \sin \theta} \left[\frac{\partial(f_r r^2 \sin \theta)}{\partial r} + \frac{\partial(f_\theta r \sin \theta)}{\partial \theta} + \frac{\partial(f_\varphi r)}{\partial \varphi} \right] \\ &= \frac{1}{r^2} \frac{\partial(r^2 f_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta f_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial f_\varphi}{\partial \varphi}. \end{aligned} \quad (0.13.7)$$

矢量场的旋度为

$$\begin{aligned}
 \nabla \times \mathbf{f} &= \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ f_1 h_1 & f_2 h_2 & f_3 h_3 \end{vmatrix} \\
 &= \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ f_r & f_\theta r & f_\varphi r \sin \theta \end{vmatrix} \\
 &= \frac{1}{r^2 \sin \theta} \left(\frac{\partial(r \sin \theta f_\varphi)}{\partial \theta} - \frac{\partial(r f_\theta)}{\partial \varphi} \right) \mathbf{e}_r + \frac{1}{r \sin \theta} \left(\frac{\partial f_r}{\partial \varphi} - \frac{\partial(r \sin \theta f_\varphi)}{\partial r} \right) \mathbf{e}_\theta \\
 &\quad + \frac{1}{r} \left(\frac{\partial(r f_\theta)}{\partial r} - \frac{\partial f_r}{\partial \theta} \right) \mathbf{e}_\varphi \\
 &= \frac{1}{r \sin \theta} \left(\frac{\partial(\sin \theta f_\varphi)}{\partial \theta} - \frac{\partial f_\theta}{\partial \varphi} \right) \mathbf{e}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial f_r}{\partial \varphi} - \frac{\partial(r f_\varphi)}{\partial r} \right) \mathbf{e}_\theta \\
 &\quad + \frac{1}{r} \left(\frac{\partial(r f_\theta)}{\partial r} - \frac{\partial f_r}{\partial \theta} \right) \mathbf{e}_\varphi. \tag{0.13.8}
 \end{aligned}$$

拉普拉斯算子为

$$\begin{aligned}
 \nabla^2 \psi &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial u_3} \right) \right] \\
 &= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{\partial \psi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{\partial}{\partial \varphi} \left(\frac{1}{\sin \theta} \frac{\partial \psi}{\partial \varphi} \right) \right] \\
 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \varphi^2}. \tag{0.13.9}
 \end{aligned}$$

总而言之，在球坐标系下，我们有

$$\nabla \psi = \frac{\partial \psi}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \varphi} \mathbf{e}_\varphi. \tag{0.13.10}$$

$$\nabla \cdot \mathbf{f} = \frac{1}{r^2} \frac{\partial(r^2 f_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta f_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial f_\varphi}{\partial \varphi}. \tag{0.13.11}$$

$$\begin{aligned}
 \nabla \times \mathbf{f} &= \frac{1}{r \sin \theta} \left(\frac{\partial(\sin \theta f_\varphi)}{\partial \theta} - \frac{\partial f_\theta}{\partial \varphi} \right) \mathbf{e}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial f_r}{\partial \varphi} - \frac{\partial(r f_\varphi)}{\partial r} \right) \mathbf{e}_\theta \\
 &\quad + \frac{1}{r} \left(\frac{\partial(r f_\theta)}{\partial r} - \frac{\partial f_r}{\partial \theta} \right) \mathbf{e}_\varphi. \tag{0.13.12}
 \end{aligned}$$

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \varphi^2}. \tag{0.13.13}$$